US 20250285494A1

(19) **United States**

(12) **Patent Application Publication**

Dua et al.

(10) **Pub. No.: US 2025/0285494 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **SYSTEMS AND METHODS FOR DYNAMIC THEME SELECTION IN ELECTRONIC GAMING**

(71) Applicant: **Aristocrat Technologies, Inc.**, Las Vegas, NV (US)

(72) Inventors: **Dinesh Dua**, Alpharetta, GA (US); **Daniel Marks**, Decatur, GA (US); **John Milliner, III**, Alpharetta, GA (US)

(21) Appl. No.: **19/219,621**

(22) Filed: **May 27, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/957,838, filed on Sep. 30, 2022, now Pat. No. 12,347,263.

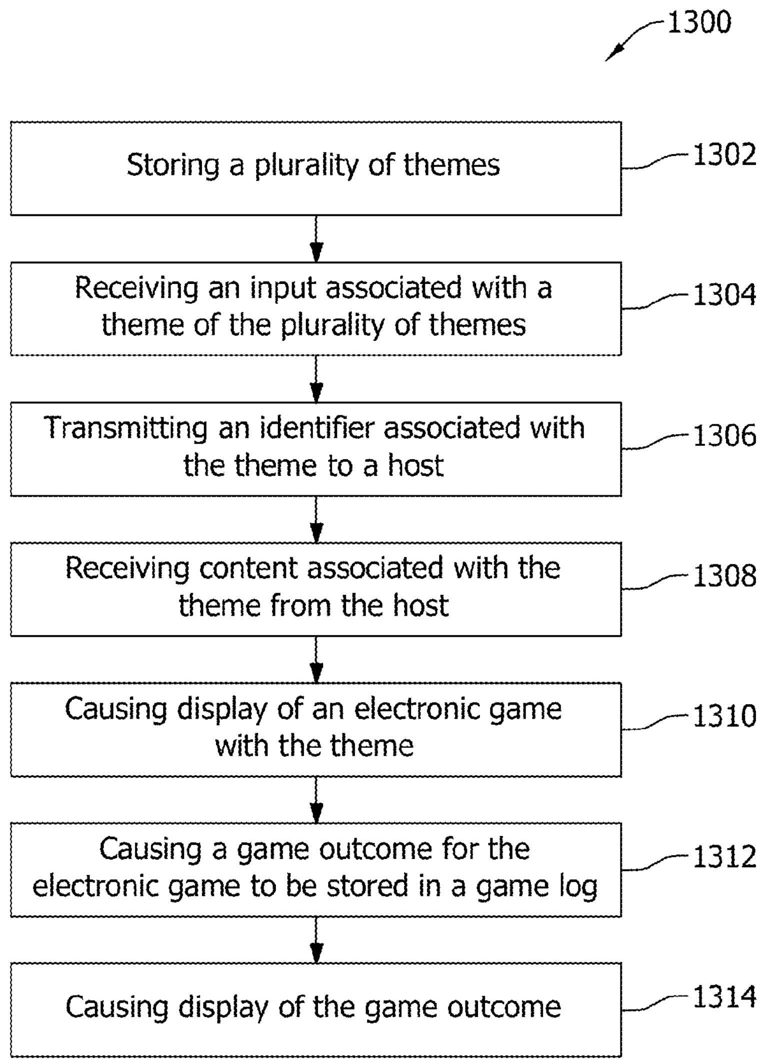
Publication Classification

(51) **Int. Cl.**
G07F 17/32 (2006.01)

(52) **U.S. Cl.**
CPC ***G07F 17/3213*** (2013.01); ***G07F 17/3209*** (2013.01); ***G07F 17/3267*** (2013.01)

(57) **ABSTRACT**

An electronic gaming system including a memory with instructions stored thereon and a processor in communication with the memory is described. The instructions, when executed by the processor, cause the processor to cause display of an electronic game with a first theme, receive an input associated with a second theme, and transmit an identifier associated with the second theme to a host component. The instructions also cause the processor to receive content associated with the second theme and cause display of the electronic game with the second theme. The instructions further cause the processor to cause a game outcome for the electronic game to be stored in a game log wherein upon generation of a replay of the game outcome the replay includes the at least one symbol, and cause display of the game outcome.



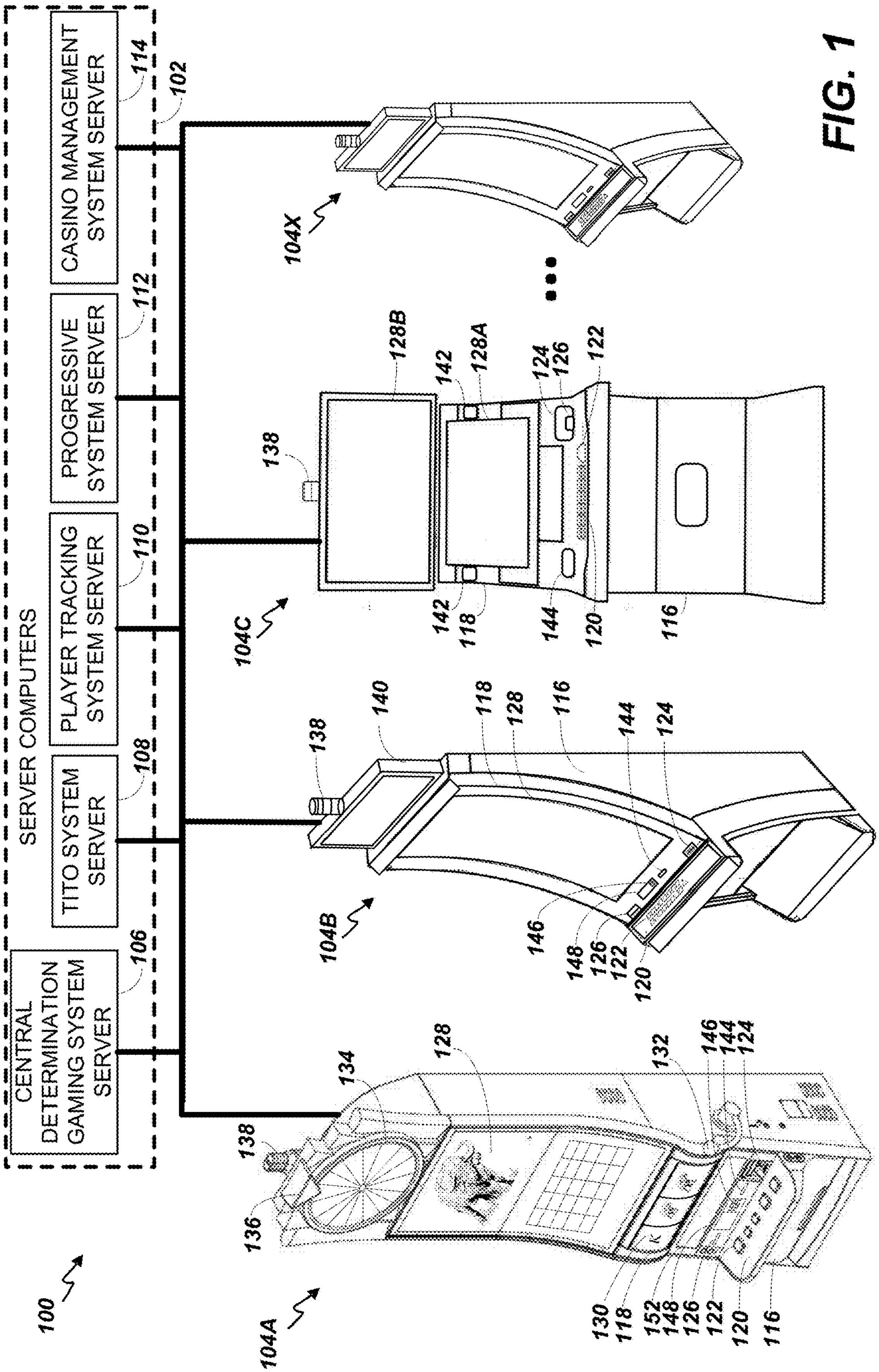

100
102
SERVER COMPUTERS
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
TITO SYSTEM SERVER
108
PLAYER TRACKING SYSTEM SERVER
110
PROGRESSIVE SYSTEM SERVER
112
CASINO MANAGEMENT SYSTEM SERVER
114
104A
104B
104C
104X
116
118
120
122
124
126
128
128A
128B
130
132
134
136
138
140
142
144
146
148
152

FIG. 1

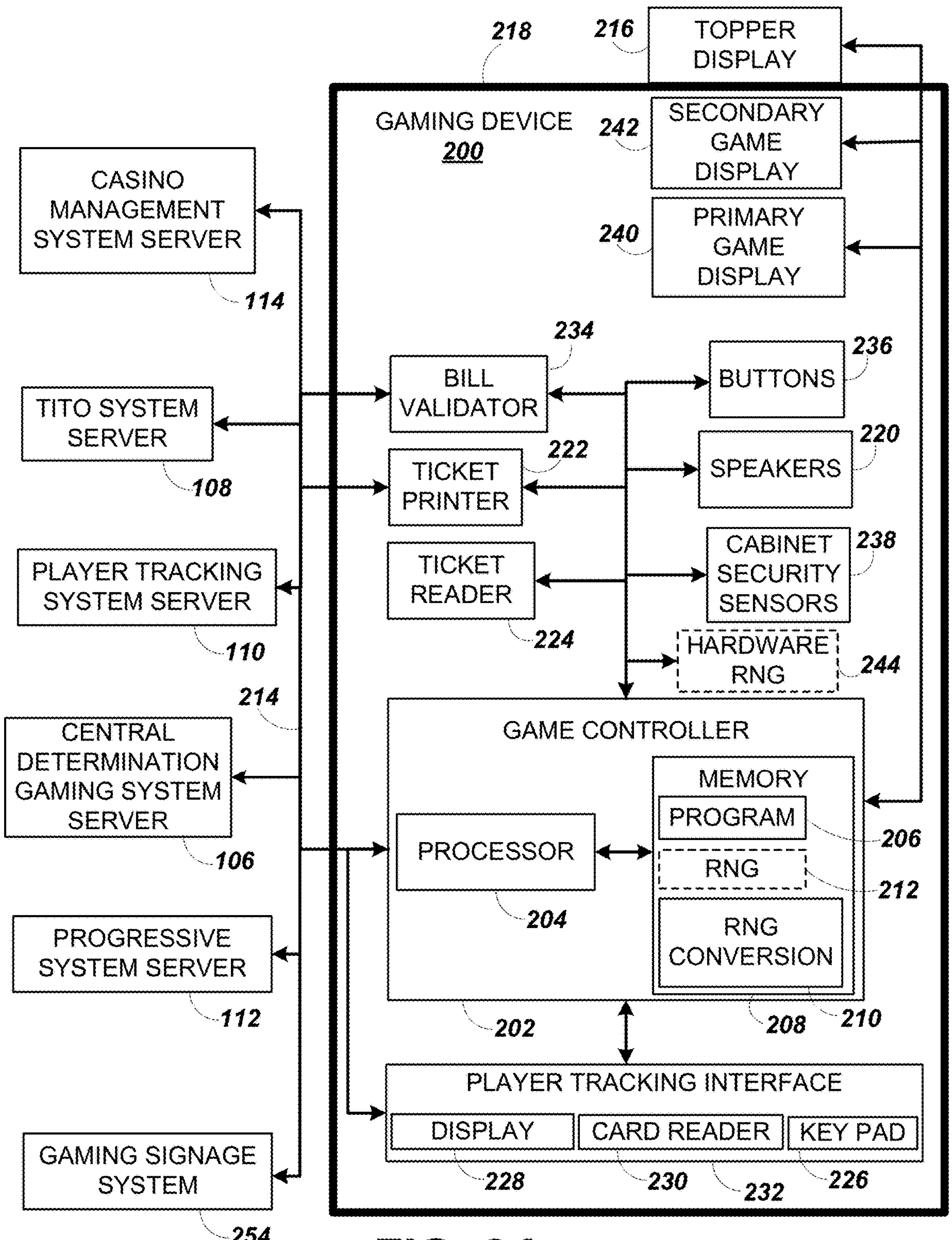

218
216
TOPPER DISPLAY
GAMING DEVICE
200
242
SECONDARY GAME DISPLAY
240
PRIMARY GAME DISPLAY
CASINO MANAGEMENT SYSTEM SERVER
114
234
BILL VALIDATOR
236
BUTTONS
TITO SYSTEM SERVER
108
222
TICKET PRINTER
220
SPEAKERS
PLAYER TRACKING SYSTEM SERVER
110
TICKET READER
224
CABINET SECURITY SENSORS
238
HARDWARE RNG
244
214
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
GAME CONTROLLER
MEMORY
PROGRAM
206
PROCESSOR
RNG
212
204
RNG CONVERSION
PROGRESSIVE SYSTEM SERVER
112
202
208
210
PLAYER TRACKING INTERFACE
DISPLAY
CARD READER
KEY PAD
GAMING SIGNAGE SYSTEM
254
228
230
232
226

FIG. 2A

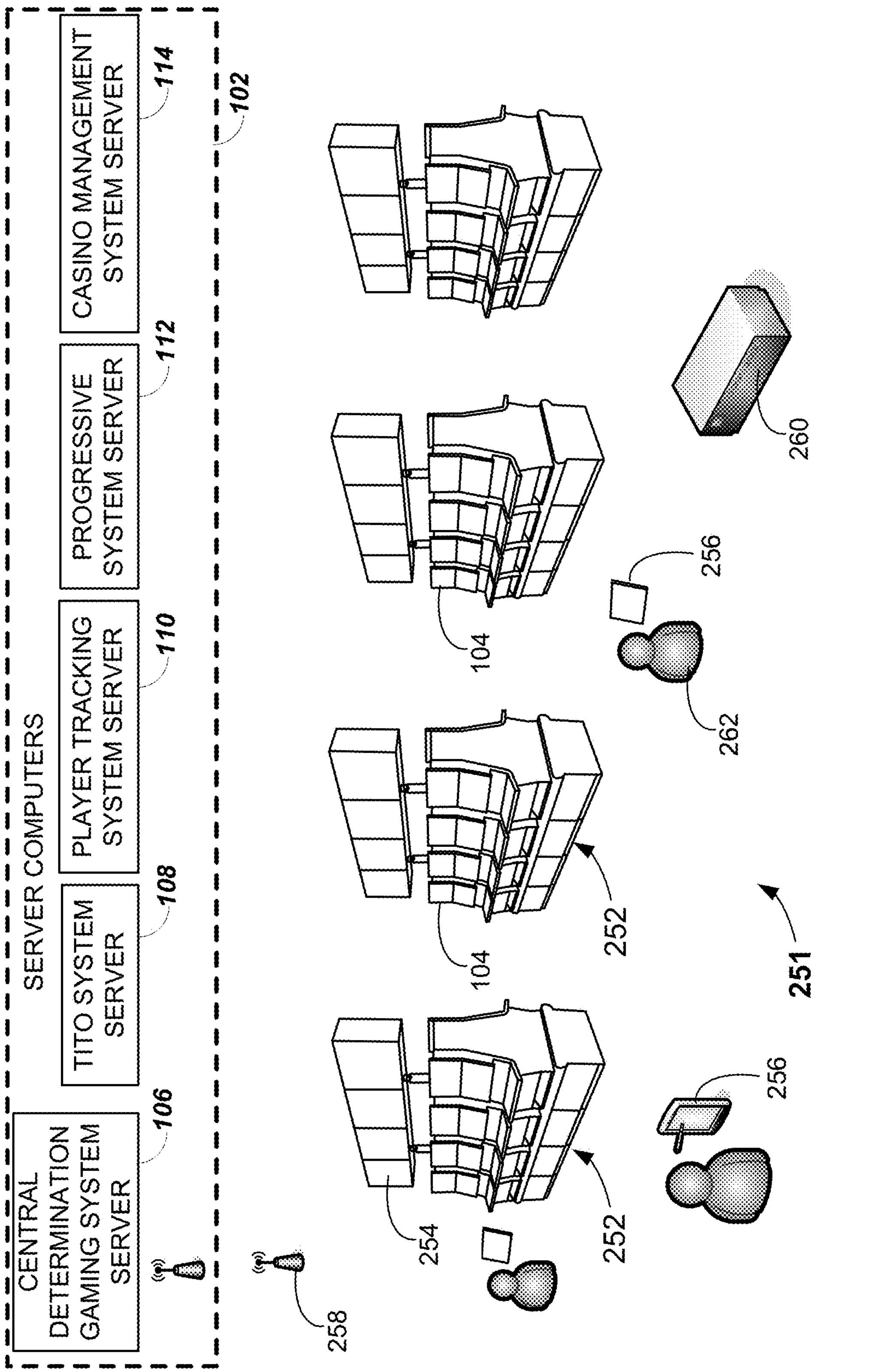
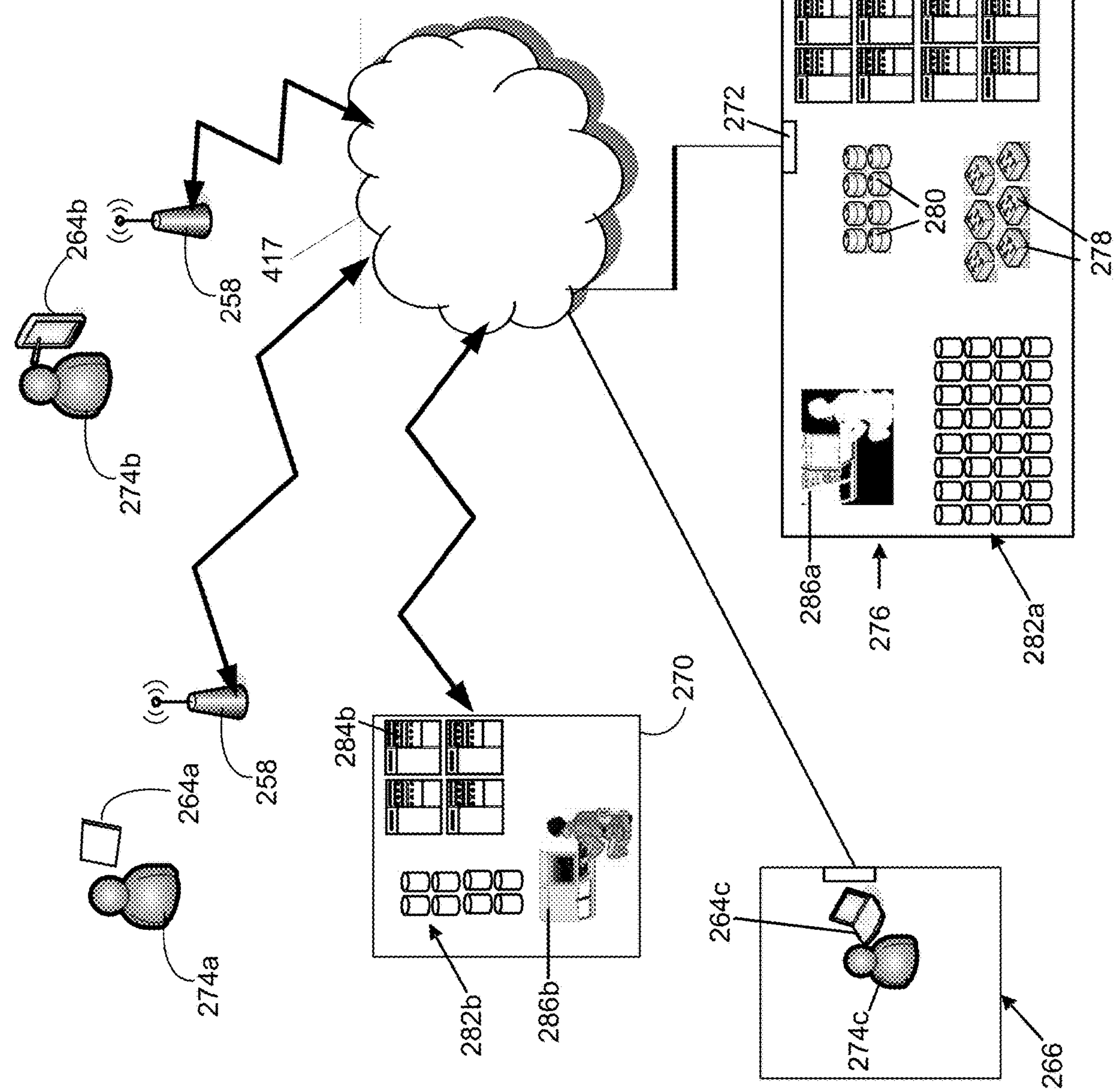

SERVER COMPUTERS
CENTRAL DETERMINATION GAMING SYSTEM SERVER
106
TITO SYSTEM SERVER
108
PLAYER TRACKING SYSTEM SERVER
110
PROGRESSIVE SYSTEM SERVER
112
CASINO MANAGEMENT SYSTEM SERVER
114
102
258
254
252
252
104
104
256
256
262
260
251

FIG. 2B

FIG. 2C


264b
274b
258
417
272
284a
280
278
286a
276
282a
264a
274a
258
270
284b
282b
286b
264c
274c
266

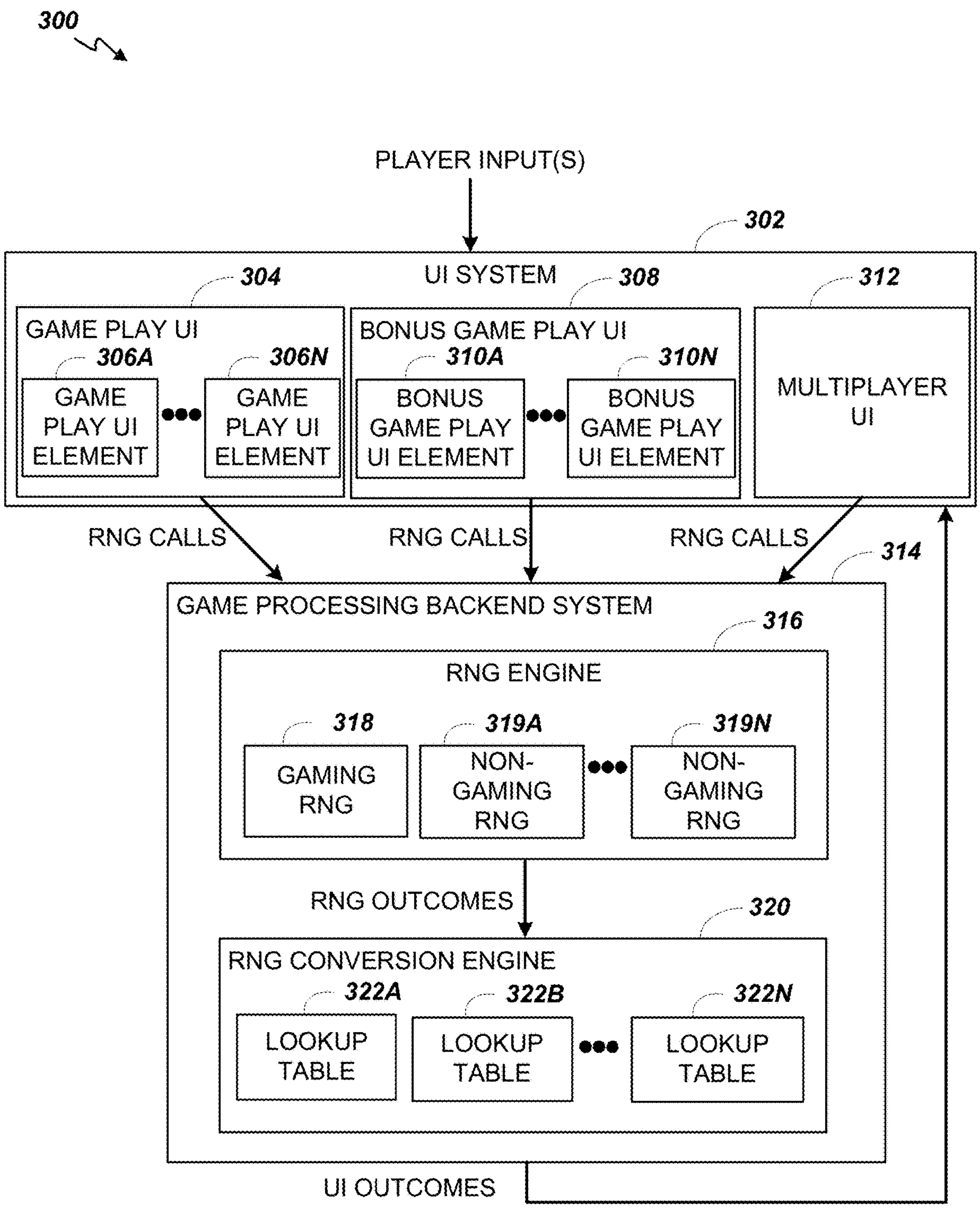

300
PLAYER INPUT(S)
302
UI SYSTEM
304
GAME PLAY UI
306A
GAME PLAY UI ELEMENT
306N
GAME PLAY UI ELEMENT
308
BONUS GAME PLAY UI
310A
BONUS GAME PLAY UI ELEMENT
310N
BONUS GAME PLAY UI ELEMENT
312
MULTIPLAYER UI
RNG CALLS
RNG CALLS
RNG CALLS
314
GAME PROCESSING BACKEND SYSTEM
316
RNG ENGINE
318
GAMING RNG
319A
NON-GAMING RNG
319N
NON-GAMING RNG
RNG OUTCOMES
320
RNG CONVERSION ENGINE
322A
LOOKUP TABLE
322B
LOOKUP TABLE
322N
LOOKUP TABLE
UI OUTCOMES

FIG. 3

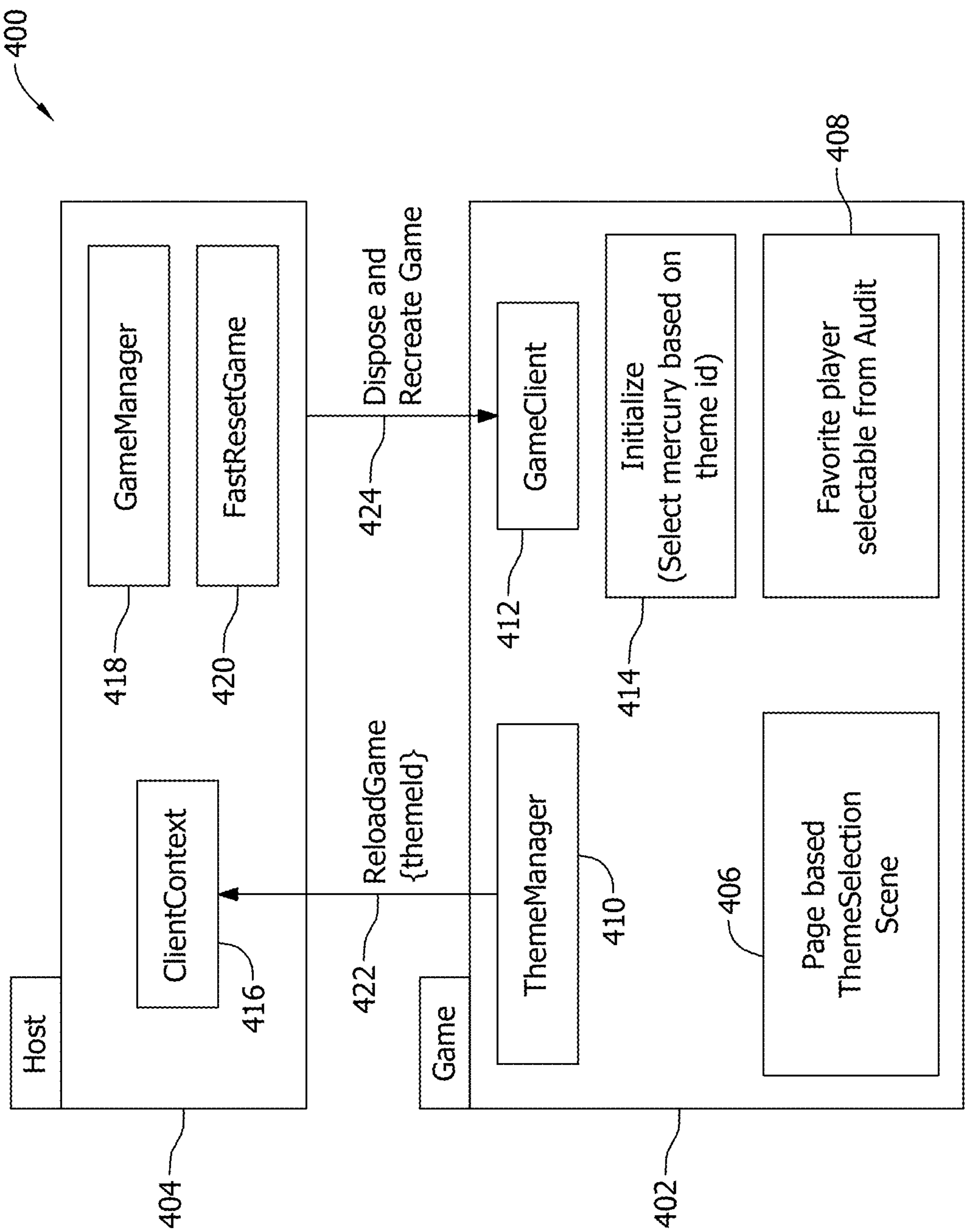

400
Host
GameManager
418
FastResetGame
420
ClientContext
416
404
Dispose and Recreate Game
424
ReloadGame {themeId}
422
Game
GameClient
412
Initialize (Select mercury based on theme id)
414
Favorite player selectable from Audit
408
ThemeManager
410
Page based ThemeSelection Scene
406
402

FIG. 4

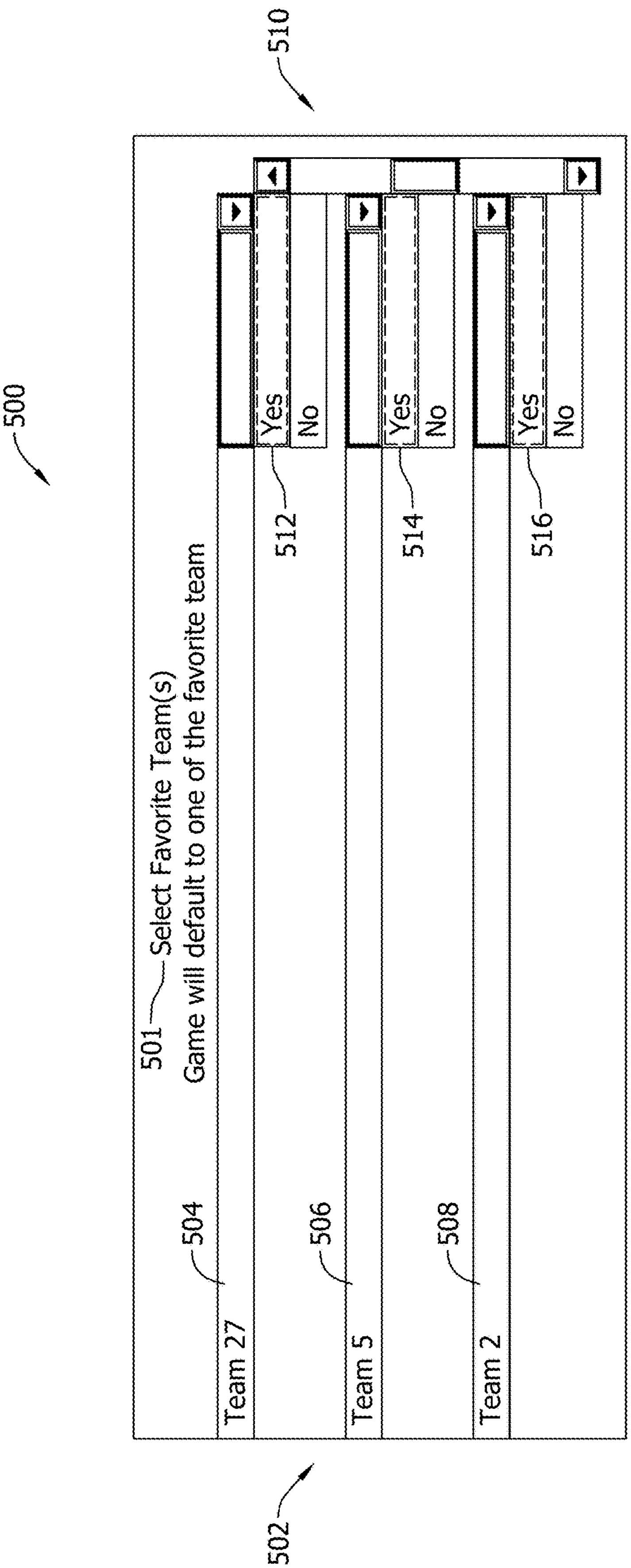

500
510
501
Select Favorite Team(s)
Game will default to one of the favorite team
Team 27
504
Yes
No
512
Team 5
506
Yes
No
514
Team 2
508
Yes
No
516
502

FIG. 5A

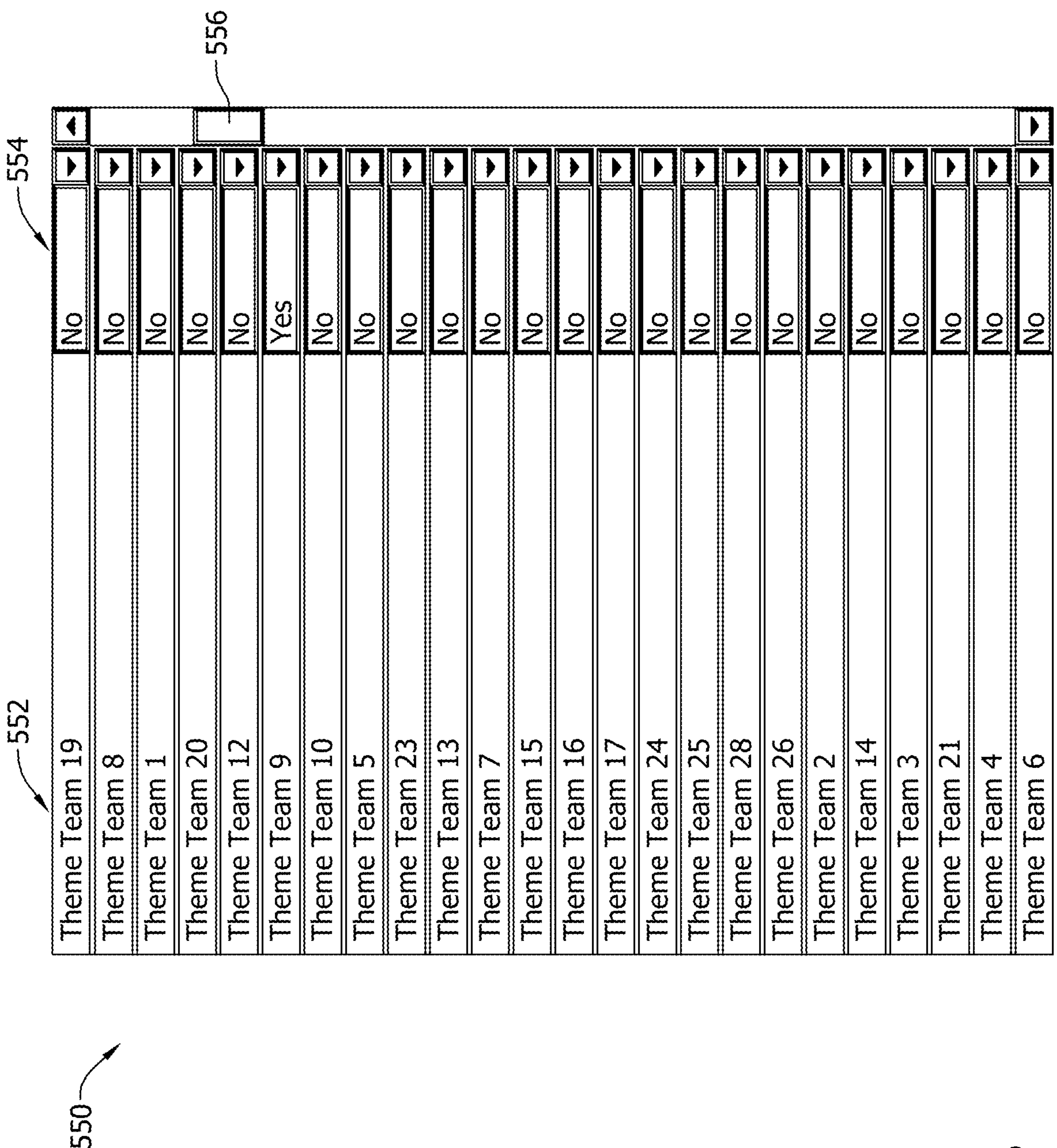

550
552
554
556
Theme Team 19 No
Theme Team 8 No
Theme Team 1 No
Theme Team 20 No
Theme Team 12 No
Theme Team 9 Yes
Theme Team 10 No
Theme Team 5 No
Theme Team 23 No
Theme Team 13 No
Theme Team 7 No
Theme Team 15 No
Theme Team 16 No
Theme Team 17 No
Theme Team 24 No
Theme Team 25 No
Theme Team 28 No
Theme Team 26 No
Theme Team 2 No
Theme Team 14 No
Theme Team 3 No
Theme Team 21 No
Theme Team 4 No
Theme Team 6 No

FIG. 5B

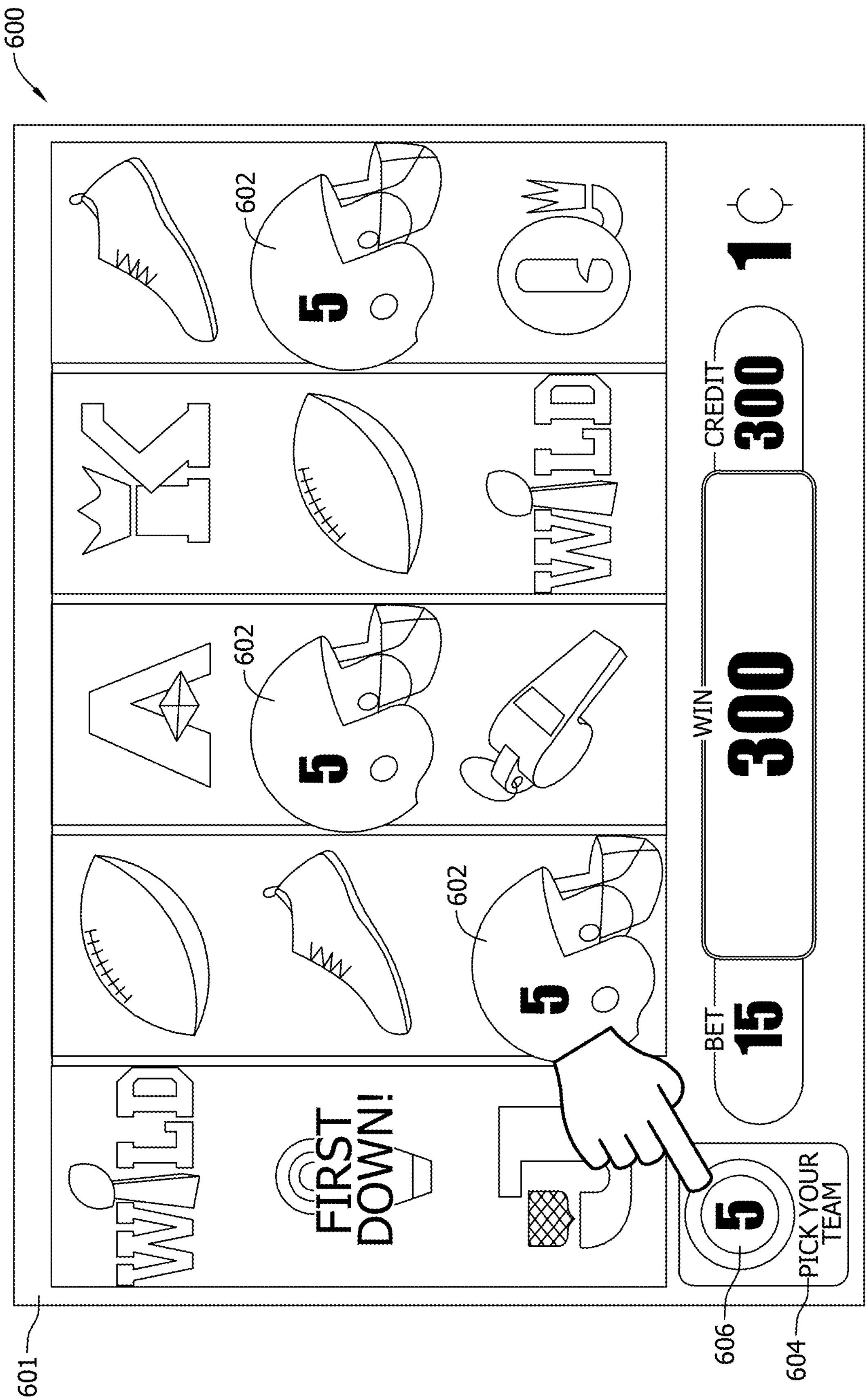

600
601
602
604
606
WILD
FIRST DOWN!
BET
15
WIN
300
CREDIT
300
1¢
PICK YOUR TEAM
5

FIG. 6

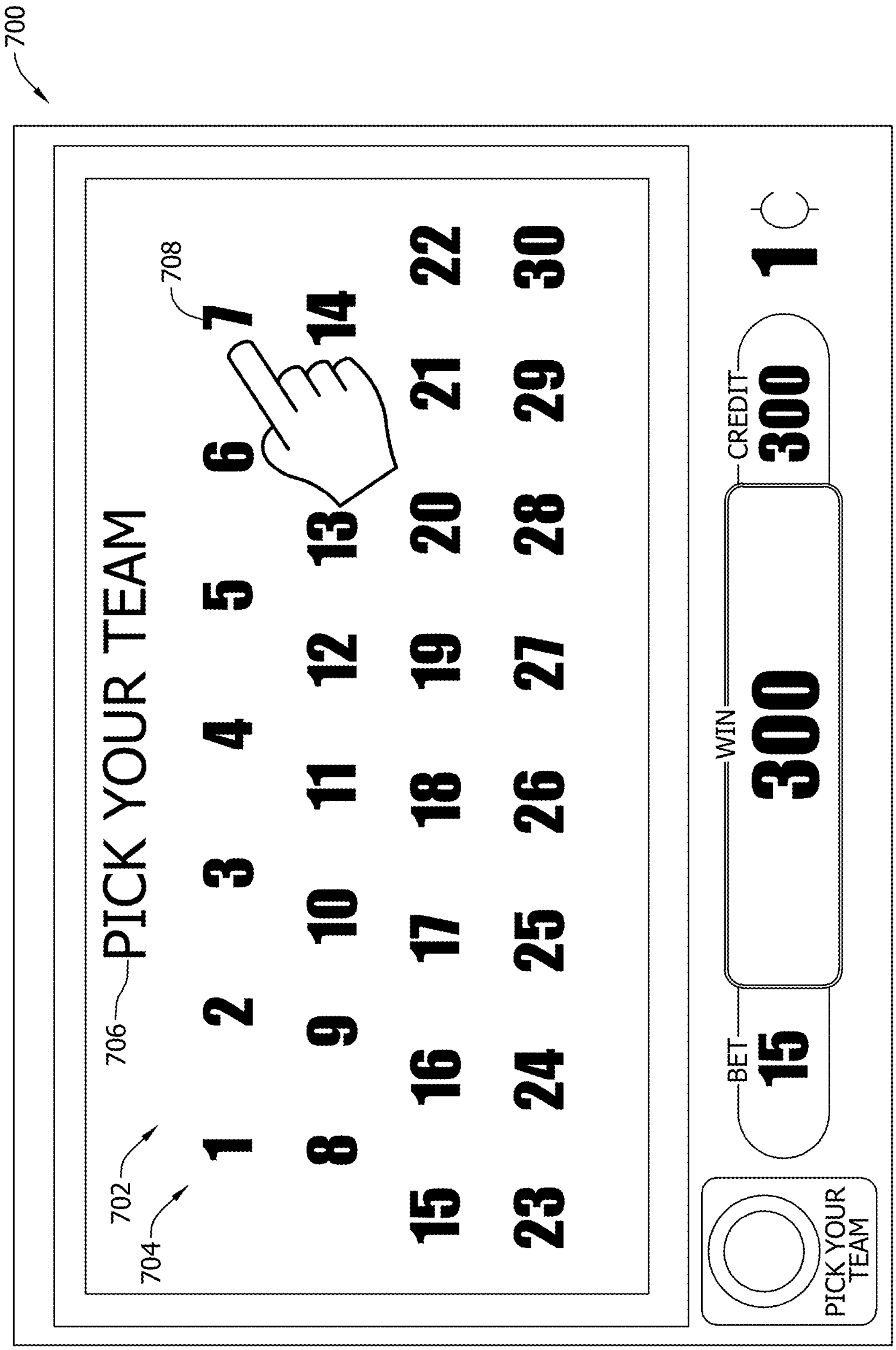

700
702
704
706
708
PICK YOUR TEAM
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21 22
23 24 25 26 27 28 29 30
BET
15
WIN
300
CREDIT
300
1¢
PICK YOUR TEAM

FIG. 7A

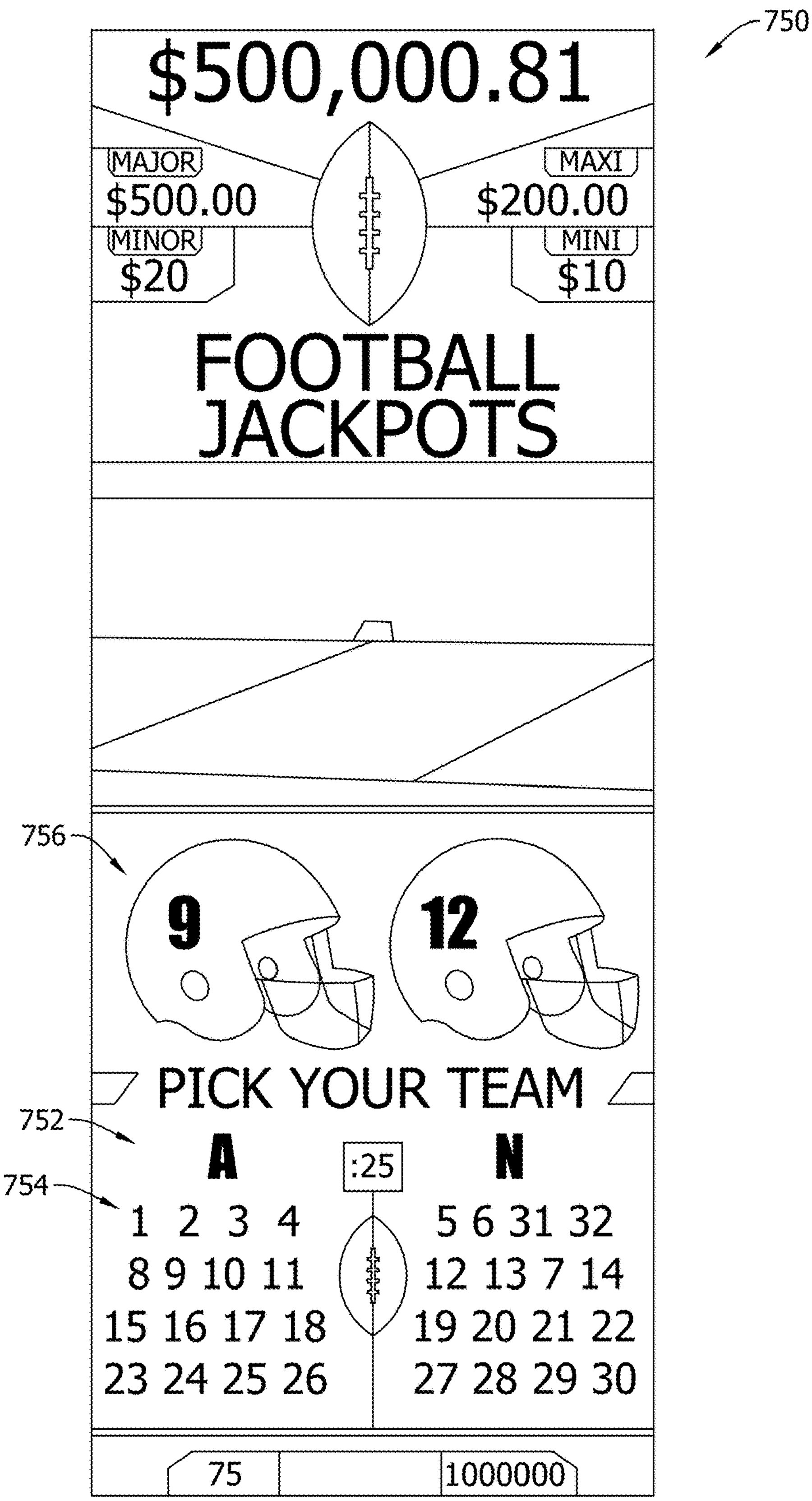

750
$500,000.81
MAJOR
$500.00
MAXI
$200.00
MINOR
$20
MINI
$10
FOOTBALL
JACKPOTS
756
9
12
PICK YOUR TEAM
752
A
:25
N
754
1 2 3 4
8 9 10 11
15 16 17 18
23 24 25 26
5 6 31 32
12 13 7 14
19 20 21 22
27 28 29 30
75
1000000

FIG. 7B

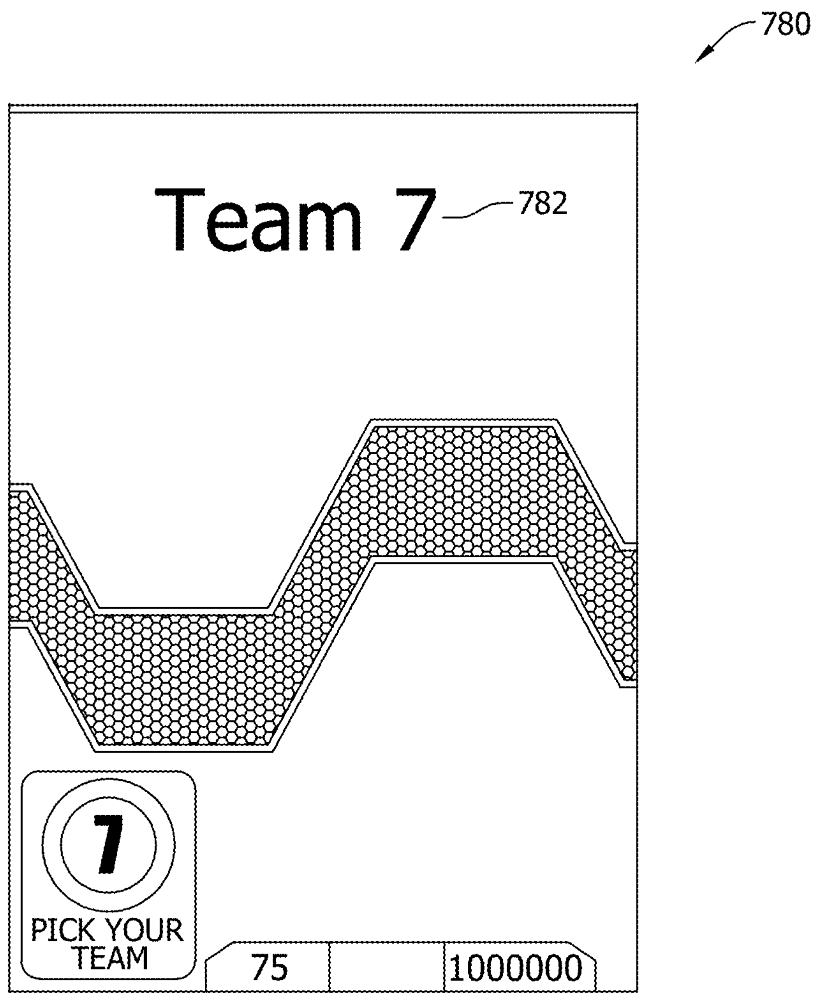


FIG. 7C

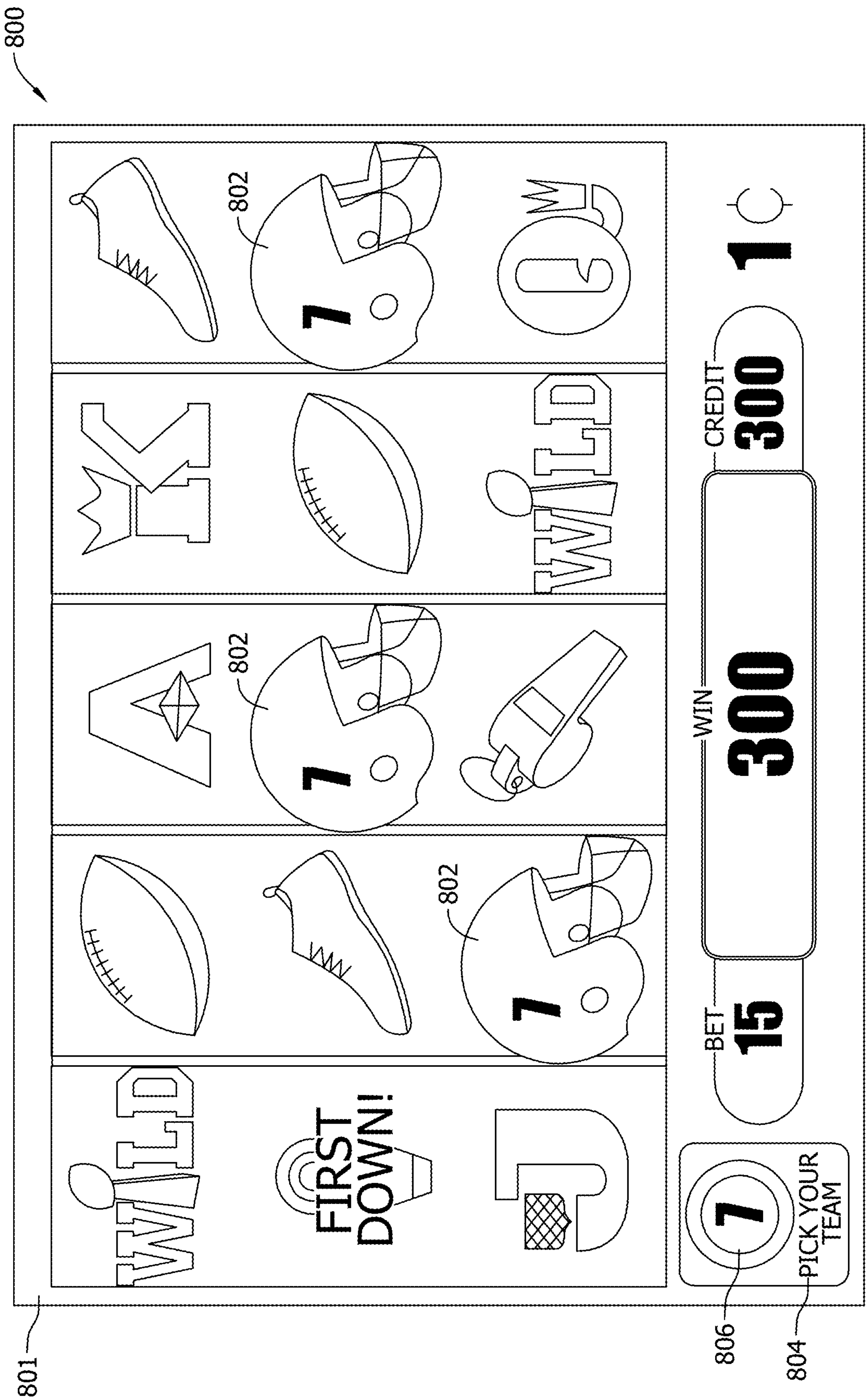

800
801
802
802
802
804
806
WILD
FIRST DOWN!
BET
15
WIN
300
CREDIT
300
1¢
PICK YOUR TEAM
7

FIG. 8A

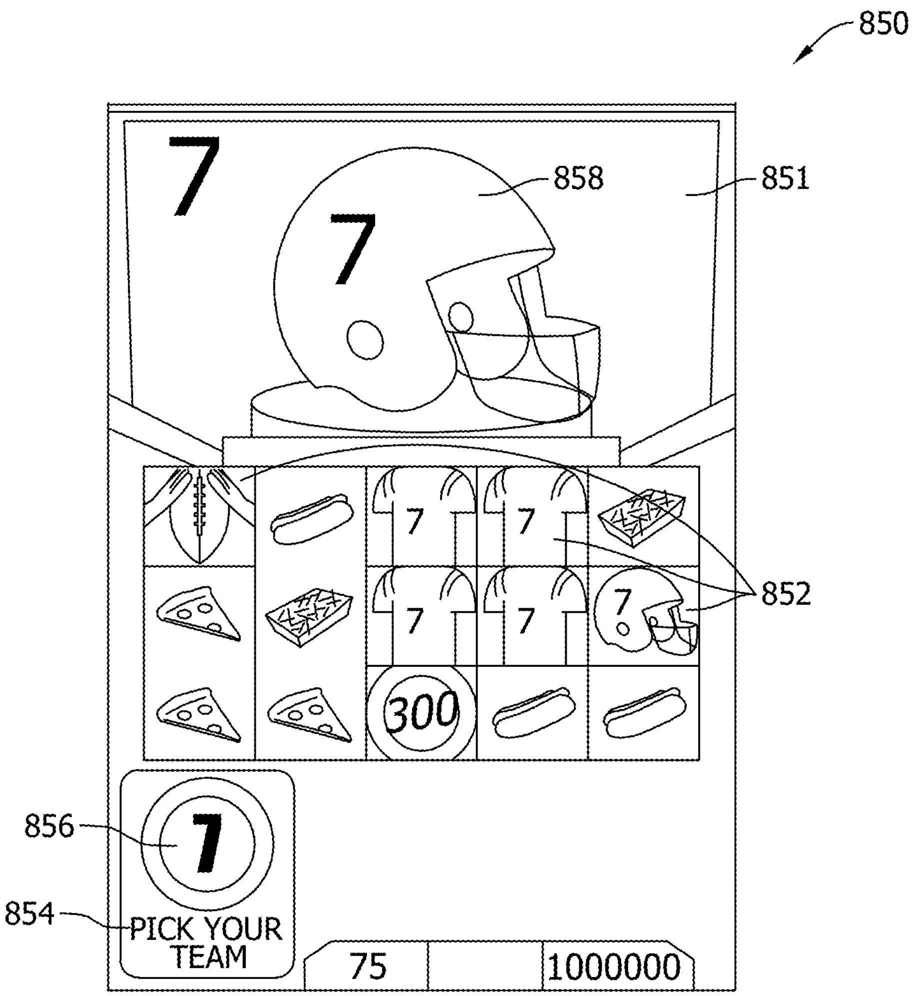

850
7
7
858
851
7
7
7
7
7
852
300
856
7
854
PICK YOUR TEAM
75
1000000

FIG. 8B

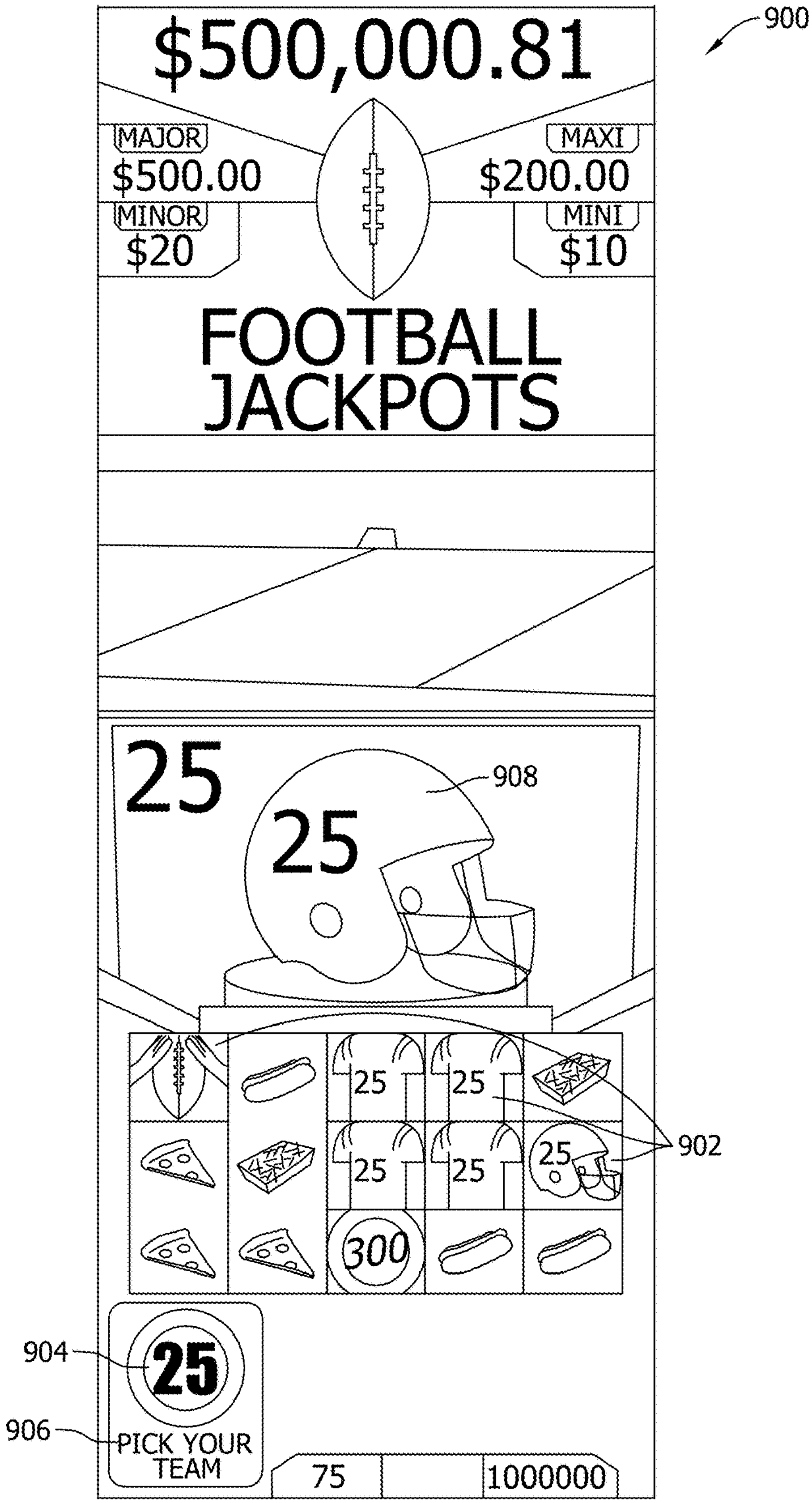

900
$500,000.81
MAJOR
$500.00
MAXI
$200.00
MINOR
$20
MINI
$10
FOOTBALL
JACKPOTS
25
25
908
25
25
25
25
25
902
300
904
25
906
PICK YOUR
TEAM
75
1000000

FIG. 9

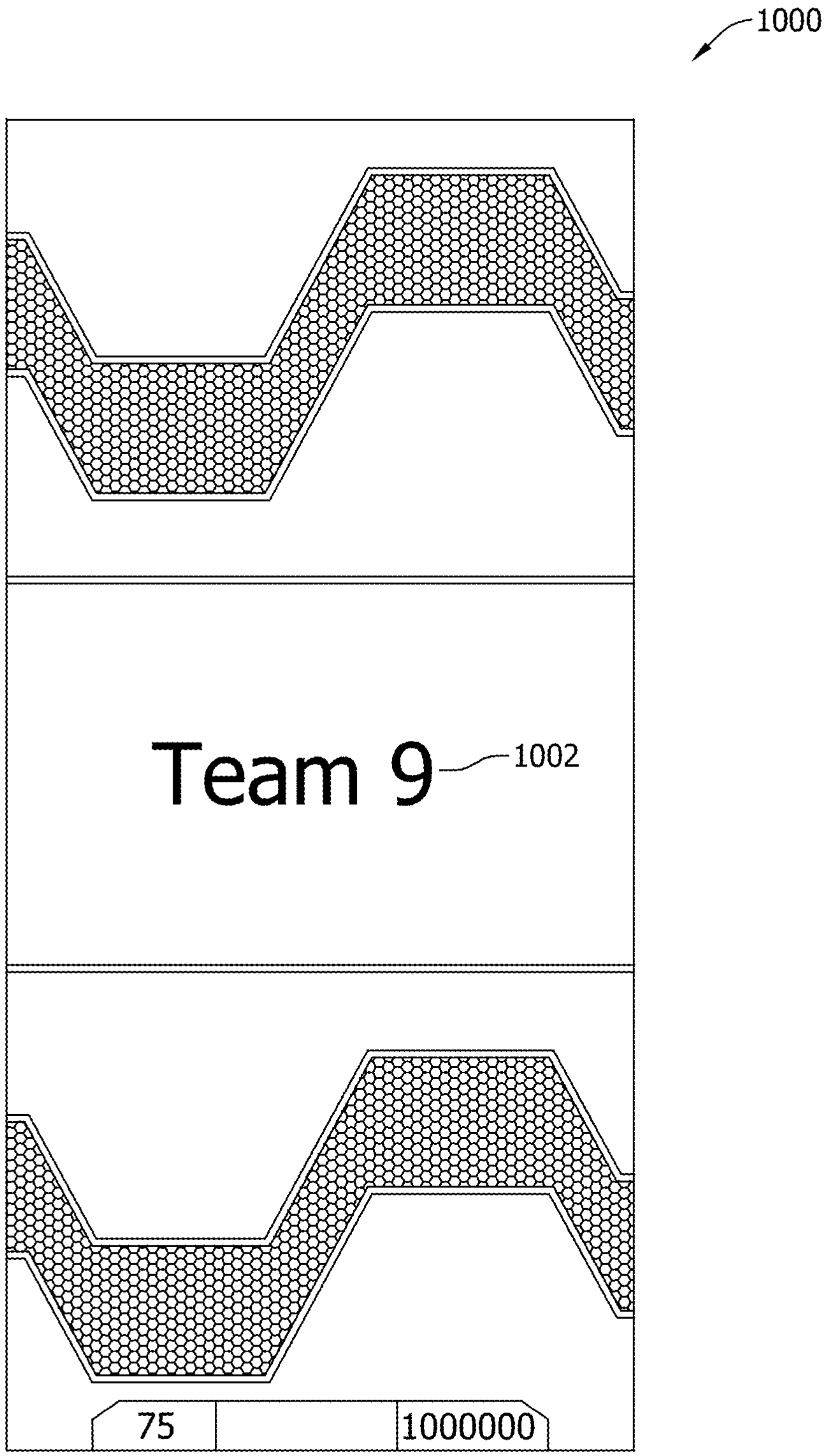


FIG. 10

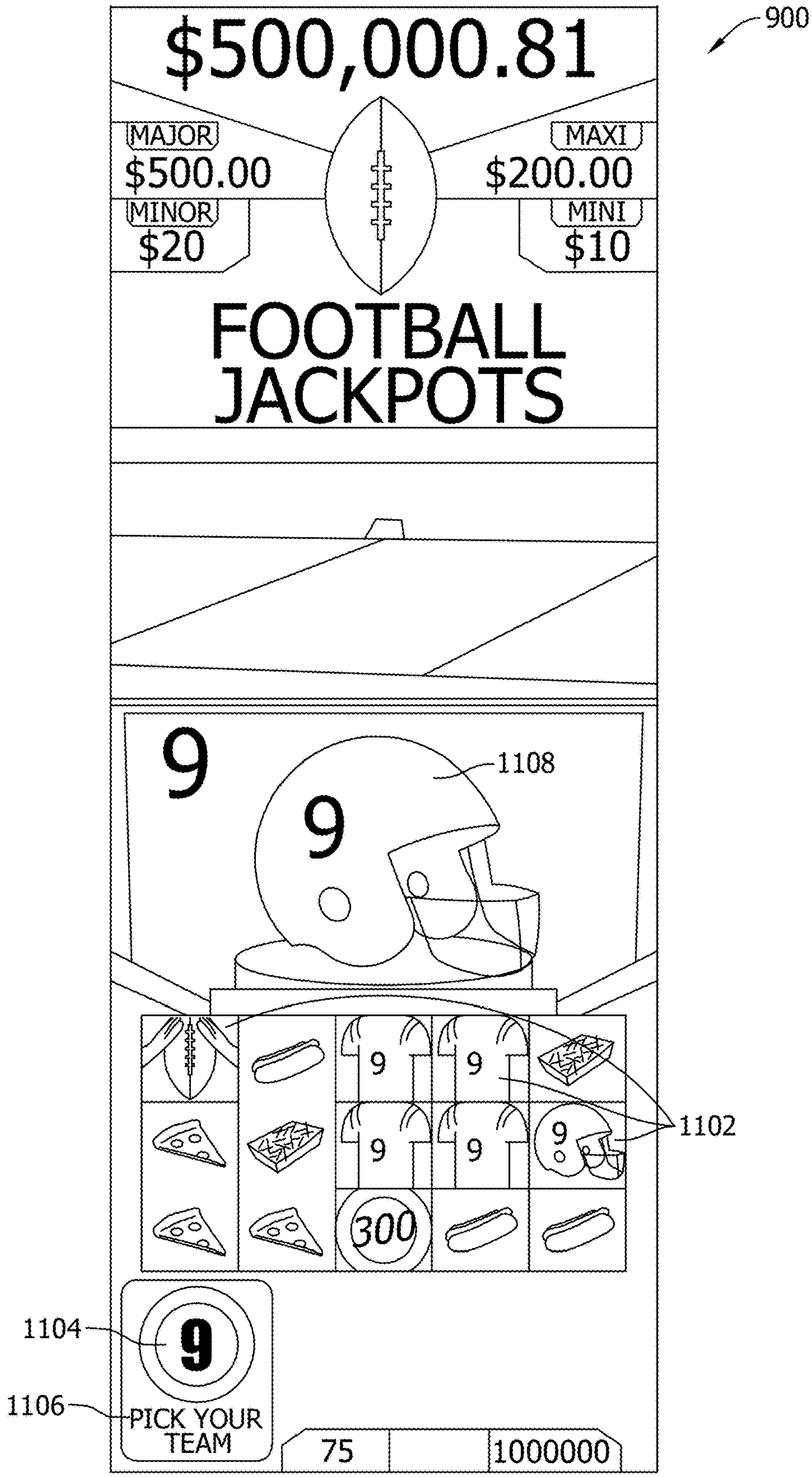

900
$500,000.81
MAJOR
$500.00
MAXI
$200.00
MINOR
$20
MINI
$10
FOOTBALL
JACKPOTS
9
9
1108
9
9
9
9
9
300
1102
1104
9
1106
PICK YOUR
TEAM
75
1000000

FIG. 11

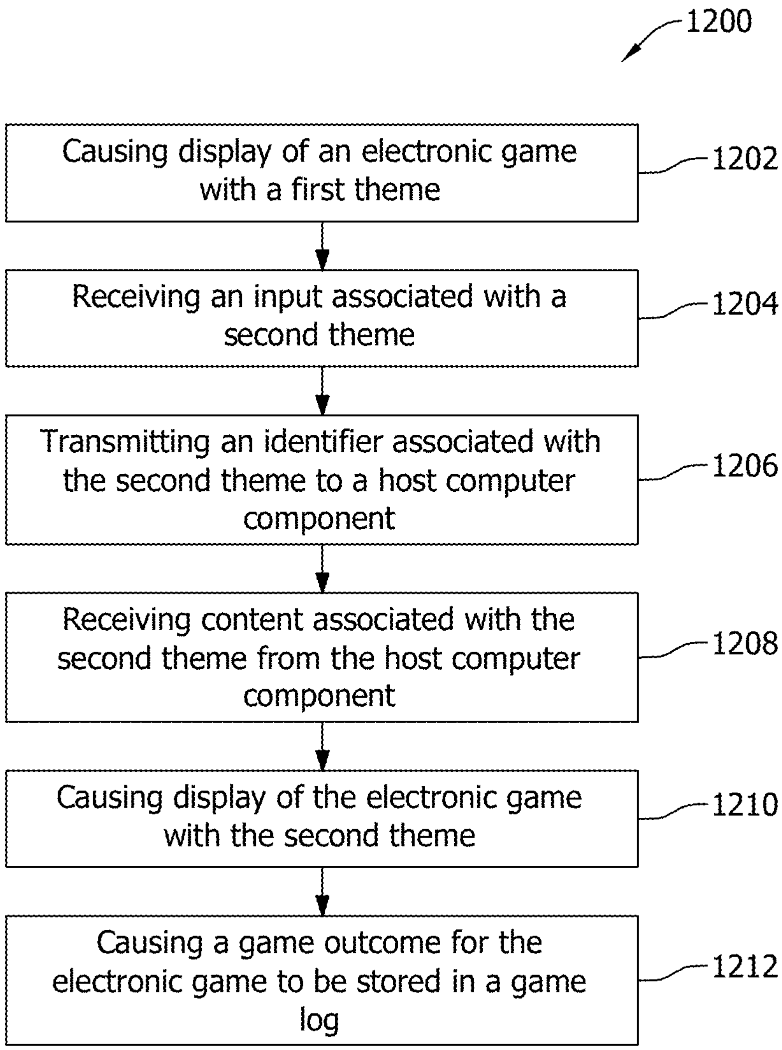

1200
Causing display of an electronic game with a first theme
1202
Receiving an input associated with a second theme
1204
Transmitting an identifier associated with the second theme to a host computer component
1206
Receiving content associated with the second theme from the host computer component
1208
Causing display of the electronic game with the second theme
1210
Causing a game outcome for the electronic game to be stored in a game log
1212

FIG. 12

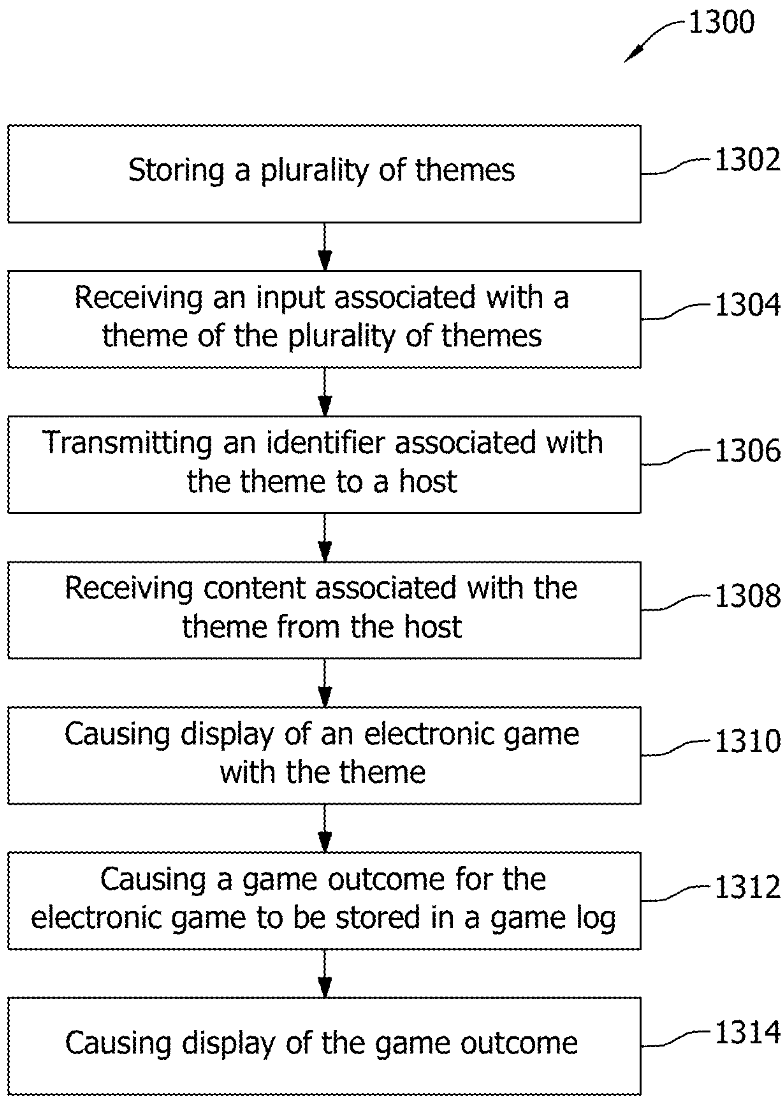

1300
Storing a plurality of themes
1302
Receiving an input associated with a theme of the plurality of themes
1304
Transmitting an identifier associated with the theme to a host
1306
Receiving content associated with the theme from the host
1308
Causing display of an electronic game with the theme
1310
Causing a game outcome for the electronic game to be stored in a game log
1312
Causing display of the game outcome
1314

FIG. 13

SYSTEMS AND METHODS FOR DYNAMIC THEME SELECTION IN ELECTRONIC GAMING

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/957,838, filed Sep. 30, 2022, which is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The field of disclosure relates generally to electronic gaming, and more specifically, to systems and methods for dynamic theme selection in electronic gaming.

BACKGROUND

[0003] Electronic gaming machines ("EGMs") or gaming devices provide a variety of wagering games such as slot games, video poker games, video blackjack games, roulette games, video bingo games, keno games and other types of games that are frequently offered at casinos and other locations. Play on EGMs typically involves a player establishing a credit balance by inputting money, or another form of monetary credit, and placing a monetary wager (from the credit balance) on one or more outcomes of an instance (or single play) of a primary or base game. In some cases, a player may qualify for a special mode of the base game, a secondary game, or a bonus round of the base game by attaining a certain winning combination or triggering event in, or related to, the base game, or after the player is randomly awarded the special mode, secondary game, or bonus round. In the special mode, secondary game, or bonus round, the player is given an opportunity to win extra game credits, game tokens or other forms of payout. In the case of "game credits" that are awarded during play, the game credits are typically added to a credit meter total on the EGM and can be provided to the player upon completion of a gaming session or when the player wants to "cash out."

[0004] "Slot" type games are often displayed to the player in the form of various symbols arrayed in a row-by-column grid or matrix. Specific matching combinations of symbols along predetermined paths (or paylines) through the matrix indicate the outcome of the game. The display typically highlights winning combinations/outcomes for identification by the player. Matching combinations and their corresponding awards are usually shown in a "pay-table" which is available to the player for reference. Often, the player may vary his/her wager to include differing numbers of paylines and/or the amount bet on each line. By varying the wager, the player may sometimes alter the frequency or number of winning combinations, frequency or number of secondary games, and/or the amount awarded.

[0005] Typical games use a random number generator (RNG) to randomly determine the outcome of each game. The game is designed to return a certain percentage of the amount wagered back to the player over the course of many plays or instances of the game, which is generally referred to as return to player (RTP). The RTP and randomness of the RNG ensure the fairness of the games and are highly regulated. Upon initiation of play, the RNG randomly determines a game outcome and symbols are then selected which correspond to that outcome. Notably, some games may include an element of skill on the part of the player and are therefore not entirely random.

BRIEF DESCRIPTION

[0006] In one aspect, an electronic gaming system is described. The electronic gaming system includes at least one memory with instructions stored thereon and at least one processor in communication with the at least one memory. The instructions, when executed by the at least one processor, cause the at least one processor to cause display of an electronic game with a first theme, the first theme including a first plurality of symbols, receive an input associated with a second theme, and transmit an identifier associated with the second theme to a host component of the at least one processor. The instructions also cause the at least one processor to receive content associated with the second theme from the host component wherein the content includes a second plurality of symbols and cause display of the electronic game with the second theme, the second theme including the second plurality of symbols. The instructions further cause the at least one processor to cause a game outcome for the electronic game to be stored in a game log in the at least one memory as being associated with the identifier wherein the game outcome includes at least one symbol of the second plurality of symbols, and wherein upon generation of a replay of the game outcome, the replay includes the at least one symbol based on the identifier and the game outcome and cause display of the game outcome.

[0007] In another aspect, at least one non-transitory, computer-readable storage medium with instructions stored thereon is described. The instructions, when executed by at least one processor, cause the at least one processor to cause display of an electronic game with a first theme, the first theme including a first plurality of symbols, receive an input associated with a second theme, and transmit an identifier associated with the second theme to a host component of the at least one processor. The instructions also cause the at least one processor to receive content associated with the second theme from the host component, wherein the content includes a second plurality of symbols, and cause display of the electronic game with the second theme, the second theme including the second plurality of symbols. The instructions further cause the at least one processor to cause a game outcome for the electronic game to be stored in a game log in at least one memory as being associated with the identifier wherein the game outcome includes at least one symbol of the second plurality of symbols, and wherein upon generation of a replay of the game outcome, the replay includes the at least one symbol based on the identifier and the game outcome and cause display of the game outcome.

[0008] In yet another aspect, an electronic gaming system is described. The electronic gaming system includes at least one memory with instructions stored thereon and at least one processor in communication with the at least one memory. The instructions, when executed by the at least one processor, cause the at least one processor to store a plurality of themes in the at least one memory, receive an input associated with a theme of the plurality of themes, and transmit an identifier associated with the theme to a host component of the at least one processor. The instructions also cause the at least one processor to receive content associated with the theme from the host component, wherein the content includes a plurality of symbols and cause display of an electronic game with the theme, the theme including the plurality of symbols. The instructions further cause the at least one processor to cause a game outcome for the electronic game to be stored in a game log in the at least one memory as being associated with the identifier wherein the game outcome includes at least one symbol of the plurality of symbols, and wherein upon generation of a replay of the game outcome, the replay includes the at least one symbol based on the identifier and the game outcome, and cause display of the game outcome.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is an exemplary diagram showing several EGMs networked with various gaming related servers.

[0010] FIG. **2**A is a block diagram showing various functional elements of an exemplary EGM.

[0011] FIG. **2**B depicts a casino gaming environment according to one example.

[0012] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure.

[0013] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture algorithm that implements a game processing pipeline for the play of a game in accordance with various implementations described herein.

[0014] FIG. **4** illustrates an example system diagram for dynamic theme selection in electronic gaming, in accordance with the present disclosure and for utilization with any of the diagrams, environments, and or architectures shown in FIGS. **1**-**3**.

[0015] FIG. **5**A illustrates an example screenshot and/or operator selection interface for dynamic theme selection in electronic gaming, continuing the example shown in FIG. **4**.

[0016] FIG. **5**B illustrates another example screenshot and/or operator selection interface for dynamic theme selection in electronic gaming, in accordance with the present disclosure.

[0017] FIG. **6** illustrates an example screenshot and/or user interface showing an output after for dynamic theme selection in electronic gaming, in accordance with the present disclosure.

[0018] FIG. **7**A illustrates an example selection screenshot and/or interface for dynamic theme selection in electronic gaming, continuing the example shown in FIG. **6**.

[0019] FIG. **7**B illustrates another example selection screenshot and/or interface for dynamic theme selection in electronic gaming, continuing the example shown in FIG. **6**.

[0020] FIG. **7**C illustrates an example transition screenshot and/or interface, continuing the examples shown in FIGS. **7**A and/or **7**B.

[0021] FIG. **8**A illustrates an example screenshot and/or interface showing an output after for dynamic theme selection in electronic gaming, based on a selection made in FIG. **7**A.

[0022] FIG. **8**B illustrates another example screenshot and/or interface showing an output after for dynamic theme selection in electronic gaming, based on a selection made in FIG. **7**B.

[0023] FIG. **9** illustrates an example screenshot and/or interface illustrating a first theme for dynamic theme selection in electronic gaming, in accordance with the present disclosure.

[0024] FIG. **10** illustrates an example screenshot and/or interface illustrating a transition to an operator-selected theme, continuing the example shown in FIG. **9**.

[0025] FIG. **11** illustrates an example screenshot and/or interface illustrating an operator-selected theme, continuing the example shown in FIG. **10**.

[0026] FIG. **12** illustrates an example method for dynamic theme selection in electronic gaming, in accordance with the present disclosure.

[0027] FIG. **13** illustrates another example method for dynamic theme selection in electronic gaming, in accordance with the present disclosure.

DETAILED DESCRIPTION

[0028] The systems and methods described herein include a gaming system that enables a user (e.g., player or operator) to dynamically select a theme for a gaming instance played on an electronic gaming device. By selecting the theme, the output of the game (e.g., the symbols displayed during game play) is dynamically altered. In the example embodiment, a selection page (e.g., or plurality of scrollable selection pages with previous, exit and next buttons) including a plurality of themes is displayed at an electronic gaming device. Upon receiving selection of a theme, the device requests a host reload a game with the selected theme (e.g., by transmitting a theme identifier (ID) associated with the selected theme to the host). In some embodiments, more than one theme may be selected (e.g., more than one theme may be implemented at a time). The host then starts a game reset cycle, recreates the game with the selected theme, and loads the game with the selected theme onto the gaming device (e.g., the game client is created again, is initialized to load the game with the selected theme).

[0029] In the example embodiment, a selected theme may be associated with symbols, animations, videos, background animations, voice overs, reel backgrounds, and/or help screens. In other words, upon selection of a theme, any of the above characteristics, or other characteristics, may be changed to be specific to the selected theme.

[0030] In some embodiments, an idle and/or default state may be configured for the game. For example, a game provider could set a default theme for the game such that before receiving a selection, the game is provided with a default theme (e.g., or the device is configured in an "attract" mode to attract players with the default theme). In some embodiments, the default theme may change dynamically (e.g., based on a schedule stored in memory). In some embodiments, the default theme may include more than one theme. In some embodiments, the default theme may be a randomly determined theme. In some embodiments, upon a game end event (e.g., a cashout event at the gaming device), the device may automatically revert to the default theme (e.g., as opposed to a theme selected for play of the game by a player).

[0031] When the game is loaded with a selected theme, the theme ID is stored and persists in a game log (e.g., that records game events, such as game outcomes, for example). The theme ID being stored in the game log supports replay (e.g., of game outcomes) and recovery (e.g., of a most recently stored theme, as explained above). Upon a recovery event, the device loads and switches to a theme associated with the most recently stored theme ID in the game log. If a theme is selected and no game is played with the selected theme, the theme ID may not be stored in the game log.

[0032] In the example embodiment, an electronic device causes display of an electronic game with a first theme including a plurality of symbols (e.g., a default theme). An input associated with a second theme is received. For example, the gaming device may cause display of a menu including a plurality of selectable icons associated with a plurality of themes and receive the input associated with the second theme at the menu. In some embodiments, themes may be selected in a number of ways different from selection at a menu including automatic selection of a theme stored in a player account based upon detection of a player associated with the player account at the gaming device (e.g., based on a card-in event and/or communication with a player device associated with the player account). In other words, the gaming device may automatically determine a player selected theme based on data associated with a player account transmitted from another device, such as a player tracking server or a player device (e.g., mobile device). Thus, the player may select a theme for storage in their player account such that upon their arrival at a gaming device, the gaming device automatically updates to correspond to the theme stored in the player account.

[0033] Based on selection of a theme, an identifier associated with the theme (e.g., a theme ID) is transmitted to a host computing component associated with the electronic game. Content associated with the second theme, including a second plurality of symbols (e.g., and/or at least one of an animation, a video, a background animation, a voice over, a reel background, or a help screen) is then received from the host.

[0034] Upon receipt of the content, the electronic game is displayed with the second theme including the second plurality of symbols. A game outcome, including at least one symbol of the second plurality of symbols, for the electronic game is generated, displayed, and stored in a game log as being associated with the identifier. Accordingly, upon replay and/or recovery of the game outcome, the replay and/or recovery includes the game outcome including the at least one symbol based on the identifier.

[0035] When a game session at the gaming device ends, the gaming device reverts back to displaying the default theme (e.g., until a player selects a different theme or the default theme is updated).

[0036] Any of the themes described herein may include a plurality of themes. Further, themes stored in memory may be updated over time (e.g., manually and/or automatically), as explained elsewhere herein.

[0037] Certain technical problems arise when implementing dynamic theme selection, as described herein. For example, game outcomes in certain gaming environments are required to be stored for, as examples, auditing purposes (e.g., audits of gameplay), replaying purposes (e.g., for a player and/or operator to replay game outcomes), and/or restoring purposes (e.g., to restore play of a game on a gaming device). Accordingly, the symbols being shown during the game need to be stored as they were shown as part of the game outcome. More variables are introduced by the ability for player selection and/or set of a default theme as described herein, and the selected theme for a game outcome needs to be loaded efficiently and stored appropriately for accurate re-creation and/or analysis of the game outcome.

[0038] Accordingly, technical solutions to at least the technical problems described herein are provided. For example, to maintain integrity of stored game outcomes, a theme ID is stored with each respective game outcome such that, upon recreation of the game outcome, the symbols displayed in the recreation match those that were shown when the game outcome was initially displayed. In other words, the game outcome is stored along with the theme ID such that, in some examples, placeholders in the game outcome are filled in with appropriate symbols that match the theme ID. Further, content associated theme may be loaded to a game client in a specific order in order to most efficiently load the content.

[0039] FIG. **1** illustrates several different models of EGMs which may be networked to various gaming related servers. Shown is a system **100** in a gaming environment including one or more server computers **102** (e.g., slot servers of a casino) that are in communication, via a communications network, with one or more gaming devices **104**A-**104**X (EGMs, slots, video poker, bingo machines, etc.) that can implement one or more aspects of the present disclosure. The gaming devices **104**A-**104**X may alternatively be portable and/or remote gaming devices such as, but not limited to, a smart phone, a tablet, a laptop, or a game console. Gaming devices **104**A-**104**X utilize specialized software and/or hardware to form non-generic, particular machines or apparatuses that comply with regulatory requirements regarding devices used for wagering or games of chance that provide monetary awards.

[0040] Communication between the gaming devices **104**A-**104**X and the server computers **102**, and among the gaming devices **104**A-**104**X, may be direct or indirect using one or more communication protocols. As an example, gaming devices **104**A-**104**X and the server computers **102** can communicate over one or more communication networks, such as over the Internet through a website maintained by a computer on a remote server or over an online data network including commercial online service providers, Internet service providers, private networks (e.g., local area networks and enterprise networks), and the like (e.g., wide area networks). The communication networks could allow gaming devices **104**A-**104**X to communicate with one another and/or the server computers **102** using a variety of communication-based technologies, such as radio frequency (RF) (e.g., wireless fidelity (WiFi®) and Bluetooth®), cable TV, satellite links and the like.

[0041] In some implementation, server computers **102** may not be necessary and/or preferred. For example, in one or more implementations, a stand-alone gaming device such as gaming device **104**A, gaming device **104**B or any of the other gaming devices **104**C-**104**X can implement one or more aspects of the present disclosure. However, it is typical to find multiple EGMs connected to networks implemented with one or more of the different server computers **102** described herein.

[0042] The server computers **102** may include a central determination gaming system server **106**, a ticket-in-ticket-out (TITO) system server **108**, a player tracking system server **110**, a progressive system server **112**, and/or a casino management system server **114**. Gaming devices **104**A-**104**X may include features to enable operation of any or all servers for use by the player and/or operator (e.g., the casino, resort, gaming establishment, tavern, pub, etc.). For example, game outcomes may be generated on a central determination gaming system server **106** and then transmitted over the network to any of a group of remote terminals or remote gaming devices **104**A-**104**X that utilize the game outcomes and display the results to the players.

[0043] Gaming device **104**A is often of a cabinet construction which may be aligned in rows or banks of similar devices for placement and operation on a casino floor. The gaming device **104**A often includes a main door which provides access to the interior of the cabinet. Gaming device **104**A typically includes a button area or button deck **120** accessible by a player that is configured with input switches or buttons **122**, an access channel for a bill validator **124**, and/or an access channel for a ticket-out printer **126**.

[0044] In FIG. **1**, gaming device **104**A is shown as a Relm XL™ model gaming device manufactured by Aristocrat® Technologies, Inc. As shown, gaming device **104**A is a reel machine having a gaming display area **118** comprising a number (typically 3 or 5) of mechanical reels **130** with various symbols displayed on them. The mechanical reels **130** are independently spun and stopped to show a set of symbols within the gaming display area **118** which may be used to determine an outcome to the game.

[0045] In many configurations, the gaming device **104**A may have a main display **128** (e.g., video display monitor) mounted to, or above, the gaming display area **118**. The main display **128** can be a high-resolution liquid crystal display (LCD), plasma, light emitting diode (LED), or organic light emitting diode (OLED) panel which may be flat or curved as shown, a cathode ray tube, or other conventional electronically controlled video monitor.

[0046] In some implementations, the bill validator **124** may also function as a "ticket-in" reader that allows the player to use a casino issued credit ticket to load credits onto the gaming device **104**A (e.g., in a cashless ticket ("TITO") system). In such cashless implementations, the gaming device **104**A may also include a "ticket-out" printer **126** for outputting a credit ticket when a "cash out" button is pressed. Cashless TITO systems are used to generate and track unique bar-codes or other indicators printed on tickets to allow players to avoid the use of bills and coins by loading credits using a ticket reader and cashing out credits using a ticket-out printer **126** on the gaming device **104**A. The gaming device **104**A can have hardware meters for purposes including ensuring regulatory compliance and monitoring the player credit balance. In addition, there can be additional meters that record the total amount of money wagered on the gaming device, total amount of money deposited, total amount of money withdrawn, total amount of winnings on gaming device **104**A.

[0047] In some implementations, a player tracking card reader **144**, a transceiver for wireless communication with a mobile device (e.g., a player's smartphone), a keypad **146**, and/or an illuminated display **148** for reading, receiving, entering, and/or displaying player tracking information is provided in gaming device **104**A. In such implementations, a game controller within the gaming device **104**A can communicate with the player tracking system server **110** to send and receive player tracking information.

[0048] Gaming device **104**A may also include a bonus topper wheel **134**. When bonus play is triggered (e.g., by a player achieving a particular outcome or set of outcomes in the primary game), bonus topper wheel **134** is operative to spin and stop with indicator arrow **136** indicating the outcome of the bonus game. Bonus topper wheel **134** is typically used to play a bonus game, but it could also be incorporated into play of the base or primary game.

[0049] A candle **138** may be mounted on the top of gaming device **104**A and may be activated by a player (e.g., using a switch or one of buttons **122**) to indicate to operations staff that gaming device **104**A has experienced a malfunction or the player requires service. The candle **138** is also often used to indicate a jackpot has been won and to alert staff that a hand payout of an award may be needed.

[0050] There may also be one or more information panels **152** which may be a back-lit, silkscreened glass panel with lettering to indicate general game information including, for example, a game denomination (e.g., $0.25 or $1), pay lines, pay tables, and/or various game related graphics. In some implementations, the information panel(s) **152** may be implemented as an additional video display.

[0051] Gaming devices **104**A have traditionally also included a handle **132** typically mounted to the side of main cabinet **116** which may be used to initiate game play.

[0052] Many or all the above described components can be controlled by circuitry (e.g., a game controller) housed inside the main cabinet **116** of the gaming device **104**A, the details of which are shown in FIG. **2**A.

[0053] An alternative example gaming device **104**B illustrated in FIG. **1** is the Arc™ model gaming device manufactured by Aristocrat® Technologies, Inc. Note that where possible, reference numerals identifying similar features of the gaming device **104**A implementation are also identified in the gaming device **104**B implementation using the same reference numbers. Gaming device **104**B does not include physical reels and instead shows game play functions on main display **128**. An optional topper screen **140** may be used as a secondary game display for bonus play, to show game features or attraction activities while a game is not in play, or any other information or media desired by the game designer or operator. In some implementations, the optional topper screen **140** may also or alternatively be used to display progressive jackpot prizes available to a player during play of gaming device **104**B.

[0054] Example gaming device **104**B includes a main cabinet **116** including a main door which opens to provide access to the interior of the gaming device **104**B. The main or service door is typically used by service personnel to refill the ticket-out printer **126** and collect bills and tickets inserted into the bill validator **124**. The main or service door may also be accessed to reset the machine, verify and/or upgrade the software, and for general maintenance operations.

[0055] Another example gaming device **104**C shown is the Helix™ model gaming device manufactured by Aristocrat® Technologies, Inc. Gaming device **104**C includes a main display **128**A that is in a landscape orientation. Although not illustrated by the front view provided, the main display **128**A may have a curvature radius from top to bottom, or alternatively from side to side. In some implementations, main display **128**A is a flat panel display. Main display **128**A is typically used for primary game play while secondary display **128**B is typically used for bonus game play, to show game features or attraction activities while the game is not in play or any other information or media desired by the game designer or operator. In some implementations, example gaming device **104**C may also include speakers **142** to output various audio such as game sound, background music, etc.

[0056] Many different types of games, including mechanical slot games, video slot games, video poker, video blackjack, video pachinko, keno, bingo, and lottery, may be provided with or implemented within the depicted gaming devices **104**A-**104**C and other similar gaming devices. Each gaming device may also be operable to provide many different games. Games may be differentiated according to themes, sounds, graphics, type of game (e.g., slot game vs. card game vs. game with aspects of skill), denomination, number of paylines, maximum jackpot, progressive or non-progressive, bonus games, and may be deployed for operation in Class **2** or Class **3**, etc.

[0057] FIG. **2**A is a block diagram depicting exemplary internal electronic components of a gaming device **200** connected to various external systems. All or parts of the gaming device **200** shown could be used to implement any one of the example gaming devices **104**A-X depicted in FIG. **1**. As shown in FIG. **2**A, gaming device **200** includes a topper display **216** or another form of a top box (e.g., a topper wheel, a topper screen, etc.) that sits above cabinet **218**. Cabinet **218** or topper display **216** may also house a number of other components which may be used to add features to a game being played on gaming device **200**, including speakers **220**, a ticket printer **222** which prints bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, a ticket reader **224** which reads bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, and a player tracking interface **232**. Player tracking interface **232** may include a keypad **226** for entering information, a player tracking display **228** for displaying information (e.g., an illuminated or video display), a card reader **230** for receiving data and/or communicating information to and from media or a device such as a smart phone enabling player tracking. FIG. **2** also depicts utilizing a ticket printer **222** to print tickets for a TITO system server **108**. Gaming device **200** may further include a bill validator **234**, player-input buttons **236** for player input, cabinet security sensors **238** to detect unauthorized opening of the cabinet **218**, a primary game display **240**, and a secondary game display **242**, each coupled to and operable under the control of game controller **202**.

[0058] The games available for play on the gaming device **200** are controlled by a game controller **202** that includes one or more processors **204**. Processor **204** represents a general-purpose processor, a specialized processor intended to perform certain functional tasks, or a combination thereof. As an example, processor **204** can be a central processing unit (CPU) that has one or more multi-core processing units and memory mediums (e.g., cache memory) that function as buffers and/or temporary storage for data. Alternatively, processor **204** can be a specialized processor, such as an application specific integrated circuit (ASIC), graphics processing unit (GPU), field-programmable gate array (FPGA), digital signal processor (DSP), or another type of hardware accelerator. In another example, processor **204** is a system on chip (SoC) that combines and integrates one or more general-purpose processors and/or one or more specialized processors. Although FIG. **2**A illustrates that game controller **202** includes a single processor **204**, game controller **202** is not limited to this representation and instead can include multiple processors **204** (e.g., two or more processors).

[0059] FIG. **2**A illustrates that processor **204** is operatively coupled to memory **208**. Memory **208** is defined herein as including volatile and nonvolatile memory and other types of non-transitory data storage components. Volatile memory is memory that do not retain data values upon loss of power. Nonvolatile memory is memory that do retain data upon a loss of power. Examples of memory **208** include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, universal serial bus (USB) flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, examples of RAM include static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), and other such devices. Examples of ROM include a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Even though FIG. **2**A illustrates that game controller **202** includes a single memory **208**, game controller **202** could include multiple memories **208** for storing program instructions and/or data.

[0060] Memory **208** can store one or more game programs **206** that provide program instructions and/or data for carrying out various implementations (e.g., game mechanics) described herein. Stated another way, game program **206** represents an executable program stored in any portion or component of memory **208**. In one or more implementations, game program **206** is embodied in the form of source code that includes human-readable statements written in a programming language or machine code that contains numerical instructions recognizable by a suitable execution system, such as a processor **204** in a game controller or other system. Examples of executable programs include: (1) a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of memory **208** and run by processor **204**; (2) source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of memory **208** and executed by processor **204**; and (3) source code that may be interpreted by another executable program to generate instructions in a random access portion of memory **208** to be executed by processor **204**.

[0061] Alternatively, game programs **206** can be set up to generate one or more game instances based on instructions and/or data that gaming device **200** exchanges with one or more remote gaming devices, such as a central determination gaming system server **106** (not shown in FIG. **2**A but shown in FIG. **1**). For purpose of this disclosure, the term "game instance" refers to a play or a round of a game that gaming device **200** presents (e.g., via a user interface (UI)) to a player. The game instance is communicated to gaming device **200** via the network **214** and then displayed on gaming device **200**. For example, gaming device **200** may execute game program **206** as video streaming software that allows the game to be displayed on gaming device **200**. When a game is stored on gaming device **200**, it may be loaded from memory **208** (e.g., from a read only memory (ROM)) or from the central determination gaming system server **106** to memory **208**.

[0062] Gaming devices, such as gaming device **200**, are highly regulated to ensure fairness and, in many cases, gaming device **200** is operable to award monetary awards (e.g., typically dispensed in the form of a redeemable voucher). Therefore, to satisfy security and regulatory requirements in a gaming environment, hardware and software architectures are implemented in gaming devices **200** that differ significantly from those of general-purpose computers. Adapting general purpose computers to function as gaming devices **200** is not simple or straightforward because of: (1) the regulatory requirements for gaming devices **200**, (2) the harsh environment in which gaming devices **200** operate, (3) security requirements, (4) fault tolerance requirements, and (5) the requirement for additional special purpose componentry enabling functionality of an EGM. These differences require substantial engineering effort with respect to game design implementation, game mechanics, hardware components, and software.

[0063] One regulatory requirement for games running on gaming device **200** generally involves complying with a certain level of randomness. Typically, gaming jurisdictions mandate that gaming devices **200** satisfy a minimum level of randomness without specifying how a gaming device **200** should achieve this level of randomness. To comply, FIG. **2**A illustrates that gaming device **200** could include an RNG **212** that utilizes hardware and/or software to generate RNG outcomes that lack any pattern. The RNG operations are often specialized and non-generic in order to comply with regulatory and gaming requirements. For example, in a slot game, game program **206** can initiate multiple RNG calls to RNG **212** to generate RNG outcomes, where each RNG call and RNG outcome corresponds to an outcome for a reel. In another example, gaming device **200** can be a Class II gaming device where RNG **212** generates RNG outcomes for creating Bingo cards. In one or more implementations, RNG **212** could be one of a set of RNGs operating on gaming device **200**. More generally, an output of the RNG **212** can be the basis on which game outcomes are determined by the game controller **202**. Game developers could vary the degree of true randomness for each RNG (e.g., pseudorandom) and utilize specific RNGs depending on game requirements. The output of the RNG **212** can include a random number or pseudorandom number (either is generally referred to as a “random number”).

[0064] In FIG. **2**A, RNG **212** and hardware RNG **244** are shown in dashed lines to illustrate that RNG **212**, hardware RNG **244**, or both can be included in gaming device **200**. In one implementation, instead of including RNG **212**, gaming device **200** could include a hardware RNG **244** that generates RNG outcomes. Analogous to RNG **212**, hardware RNG **244** performs specialized and non-generic operations in order to comply with regulatory and gaming requirements. For example, because of regulation requirements, hardware RNG **244** could be a random number generator that securely produces random numbers for cryptography use. The gaming device **200** then uses the secure random numbers to generate game outcomes for one or more game features. In another implementation, the gaming device **200** could include both hardware RNG **244** and RNG **212**. RNG **212** may utilize the RNG outcomes from hardware RNG **244** as one of many sources of entropy for generating secure random numbers for the game features.

[0065] Another regulatory requirement for running games on gaming device **200** includes ensuring a certain level of RTP. Similar to the randomness requirement discussed above, numerous gaming jurisdictions also mandate that gaming device **200** provides a minimum level of RTP (e.g., RTP of at least 75%). A game can use one or more lookup tables (also called weighted tables) as part of a technical solution that satisfies regulatory requirements for randomness and RTP. In particular, a lookup table can integrate game features (e.g., trigger events for special modes or bonus games; newly introduced game elements such as extra reels, new symbols, or new cards; stop positions for dynamic game elements such as spinning reels, spinning wheels, or shifting reels; or card selections from a deck) with random numbers generated by one or more RNGs, so as to achieve a given level of volatility for a target level of RTP. (In general, volatility refers to the frequency or probability of an event such as a special mode, payout, etc. For example, for a target level of RTP, a higher-volatility game may have a lower payout most of the time with an occasional bonus having a very high payout, while a lower-volatility game has a steadier payout with more frequent bonuses of smaller amounts.) Configuring a lookup table can involve engineering decisions with respect to how RNG outcomes are mapped to game outcomes for a given game feature, while still satisfying regulatory requirements for RTP. Configuring a lookup table can also involve engineering decisions about whether different game features are combined in a given entry of the lookup table or split between different entries (for the respective game features), while still satisfying regulatory requirements for RTP and allowing for varying levels of game volatility.

[0066] FIG. **2**A illustrates that gaming device **200** includes an RNG conversion engine **210** that translates the RNG outcome from RNG **212** to a game outcome presented to a player. To meet a designated RTP, a game developer can set up the RNG conversion engine **210** to utilize one or more lookup tables to translate the RNG outcome to a symbol element, stop position on a reel strip layout, and/or randomly chosen aspect of a game feature. As an example, the lookup tables can regulate a prize payout amount for each RNG outcome and how often the gaming device **200** pays out the prize payout amounts. The RNG conversion engine **210** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. The mapping between the RNG outcome to the game outcome controls the frequency in hitting certain prize payout amounts.

[0067] FIG. **2**A also depicts that gaming device **200** is connected over network **214** to player tracking system server **110**. Player tracking system server **110** may be, for example, an OASIS® system manufactured by Aristocrat® Technologies, Inc. Player tracking system server **110** is used to track play (e.g. amount wagered, games played, time of play and/or other quantitative or qualitative measures) for individual players so that an operator may reward players in a loyalty program. The player may use the player tracking interface **232** to access his/her account information, activate free play, and/or request various information. Player tracking or loyalty programs seek to reward players for their play and help build brand loyalty to the gaming establishment. The rewards typically correspond to the player’s level of patronage (e.g., to the player’s playing frequency and/or total amount of game plays at a given casino). Player tracking rewards may be complimentary and/or discounted meals, lodging, entertainment and/or additional play. Player tracking information may be combined with other information that is now readily obtainable by a casino management system.

[0068] When a player wishes to play the gaming device **200**, he/she can insert cash or a ticket voucher through a coin acceptor (not shown) or bill validator **234** to establish a credit balance on the gaming device. The credit balance is used by the player to place wagers on instances of the game and to receive credit awards based on the outcome of winning instances. The credit balance is decreased by the amount of each wager and increased upon a win. The player can add additional credits to the balance at any time. The player may also optionally insert a loyalty club card into the card reader **230**. During the game, the player views with one or more UIs, the game outcome on one or more of the primary game display **240** and secondary game display **242**. Other game and prize information may also be displayed.

[0069] For each game instance, a player may make selections, which may affect play of the game. For example, the player may vary the total amount wagered by selecting the amount bet per line and the number of lines played. In many games, the player is asked to initiate or select options during course of game play (such as spinning a wheel to begin a bonus round or select various items during a feature game). The player may make these selections using the player-input buttons **236**, the primary game display **240** which may be a touch screen, or using some other device which enables a player to input information into the gaming device **200**.

[0070] During certain game events, the gaming device **200** may display visual and auditory effects that can be perceived by the player. These effects add to the excitement of a game, which makes a player more likely to enjoy the playing experience. Auditory effects include various sounds that are projected by the speakers **220**. Visual effects include flashing lights, strobing lights or other patterns displayed from lights on the gaming device **200** or from lights behind the information panel **152** (FIG. **1**).

[0071] When the player is done, he/she cashes out the credit balance (typically by pressing a cash out button to receive a ticket from the ticket printer **222**). The ticket may be "cashed-in" for money or inserted into another machine to establish a credit balance for play.

[0072] Additionally, or alternatively, gaming devices **104**A-**104**X and **200** can include or be coupled to one or more wireless transmitters, receivers, and/or transceivers (not shown in FIGS. **1** and **2**A) that communicate (e.g., Bluetooth® or other near-field communication technology) with one or more mobile devices to perform a variety of wireless operations in a casino environment. Examples of wireless operations in a casino environment include detecting the presence of mobile devices, performing credit, points, comps, or other marketing or hard currency transfers, establishing wagering sessions, and/or providing a personalized casino-based experience using a mobile application. In one implementation, to perform these wireless operations, a wireless transmitter or transceiver initiates a secure wireless connection between a gaming device **104**A-**104**X and **200** and a mobile device. After establishing a secure wireless connection between the gaming device **104**A-**104**X and **200** and the mobile device, the wireless transmitter or transceiver does not send and/or receive application data to and/or from the mobile device. Rather, the mobile device communicates with gaming devices **104**A-**104**X and **200** using another wireless connection (e.g., WiFi® or cellular network). In another implementation, a wireless transceiver establishes a secure connection to directly communicate with the mobile device. The mobile device and gaming device **104**A-**104**X and **200** sends and receives data utilizing the wireless transceiver instead of utilizing an external network. For example, the mobile device would perform digital wallet transactions by directly communicating with the wireless transceiver. In one or more implementations, a wireless transmitter could broadcast data received by one or more mobile devices without establishing a pairing connection with the mobile devices.

[0073] Although FIGS. **1** and **2**A illustrate specific implementations of a gaming device (e.g., gaming devices **104**A-**104**X and **200**), the disclosure is not limited to those implementations shown in FIGS. **1** and **2**. For example, not all gaming devices suitable for implementing implementations of the present disclosure necessarily include top wheels, top boxes, information panels, cashless ticket systems, and/or player tracking systems. Further, some suitable gaming devices have only a single game display that includes only a mechanical set of reels and/or a video display, while others are designed for bar counters or tabletops and have displays that face upwards. Gaming devices **104**A-**104**X and **200** may also include other processors that are not separately shown. Using FIG. **2**A as an example, gaming device **200** could include display controllers (not shown in FIG. **2**A) configured to receive video input signals or instructions to display images on game displays **240** and **242**. Alternatively, such display controllers may be integrated into the game controller **202**. The use and discussion of FIGS. **1** and **2** are examples to facilitate ease of description and explanation.

[0074] FIG. **2**B depicts a casino gaming environment according to one example. In this example, the casino **251** includes banks **252** of EGMs **104**. In this example, each bank **252** of EGMs **104** includes a corresponding gaming signage system **254** (also shown in FIG. **2**A). According to this implementation, the casino **251** also includes mobile gaming devices **256**, which are also configured to present wagering games in this example. The mobile gaming devices **256** may, for example, include tablet devices, cellular phones, smart phones and/or other handheld devices. In this example, the mobile gaming devices **256** are configured for communication with one or more other devices in the casino **251**, including but not limited to one or more of the server computers **102**, via wireless access points **258**.

[0075] According to some examples, the mobile gaming devices **256** may be configured for stand-alone determination of game outcomes. However, in some alternative implementations the mobile gaming devices **256** may be configured to receive game outcomes from another device, such as the central determination gaming system server **106**, one of the EGMs **104**, etc.

[0076] Some mobile gaming devices **256** may be configured to accept monetary credits from a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, via a patron casino account, etc. However, some mobile gaming devices **256** may not be configured to accept monetary credits via a credit or debit card. Some mobile gaming devices **256** may include a ticket reader and/or a ticket printer whereas some mobile gaming devices **256** may not, depending on the particular implementation.

[0077] In some implementations, the casino **251** may include one or more kiosks **260** that are configured to facilitate monetary transactions involving the mobile gaming devices **256**, which may include cash out and/or cash in transactions. The kiosks **260** may be configured for wired and/or wireless communication with the mobile gaming devices **256**. The kiosks **260** may be configured to accept monetary credits from casino patrons **262** and/or to dispense monetary credits to casino patrons **262** via cash, a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, etc. According to some examples, the kiosks **260** may be configured to accept monetary credits from a casino patron and to provide a corresponding amount of monetary credits to a mobile gaming device **256** for wagering purposes, e.g., via a wireless link such as a near-field communications link. In some such examples, when a casino patron **262** is ready to cash out, the casino patron **262** may select a cash out option provided by a mobile gaming device **256**, which may include a real button or a virtual button (e.g., a button provided via a graphical user interface) in some instances. In some such examples, the mobile gaming device **256** may send a "cash out" signal to a kiosk **260** via a wireless link in response to receiving a "cash out" indication from a casino patron. The kiosk **260** may provide monetary credits to the casino patron **262** corresponding to the "cash out" signal, which may be in the form of cash, a credit ticket, a credit transmitted to a financial account corresponding to the casino patron, etc.

[0078] In some implementations, a cash-in process and/or a cash-out process may be facilitated by the TITO system server **108**. For example, the TITO system server **108** may control, or at least authorize, ticket-in and ticket-out transactions that involve a mobile gaming device **256** and/or a kiosk **260**.

[0079] Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information. For example, some mobile gaming devices **256** may be configured for wireless communication with the player tracking system server **110**. Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information via wireless communication with a patron's player loyalty card, a patron's smartphone, etc.

[0080] According to some implementations, a mobile gaming device **256** may be configured to provide safeguards that prevent the mobile gaming device **256** from being used by an unauthorized person. For example, some mobile gaming devices **256** may include one or more biometric sensors and may be configured to receive input via the biometric sensor(s) to verify the identity of an authorized patron. Some mobile gaming devices **256** may be configured to function only within a predetermined or configurable area, such as a casino gaming area.

[0081] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure. As with other figures presented in this disclosure, the numbers, types and arrangements of gaming devices shown in FIG. **2**C are merely shown by way of example. In this example, various gaming devices, including but not limited to end user devices (EUDs) **264***a*, **264***b* and **264***c* are capable of communication via one or more networks **417**. The networks **417** may, for example, include one or more cellular telephone networks, the Internet, etc. In this example, the EUDs **264***a* and **264***b* are mobile devices: according to this example the EUD **264***a* is a tablet device and the EUD **264***b* is a smart phone. In this implementation, the EUD **264***c* is a laptop computer that is located within a residence **266** at the time depicted in FIG. **2**C. Accordingly, in this example the hardware of EUDs is not specifically configured for online gaming, although each EUD is configured with software for online gaming. For example, each EUD may be configured with a web browser. Other implementations may include other types of EUD, some of which may be specifically configured for online gaming.

[0082] In this example, a gaming data center **276** includes various devices that are configured to provide online wagering games via the networks **417**. The gaming data center **276** is capable of communication with the networks **417** via the gateway **272**. In this example, switches **278** and routers **280** are configured to provide network connectivity for devices of the gaming data center **276**, including storage devices **282***a*, servers **284***a* and one or more workstations **286***a*. The servers **284***a* may, for example, be configured to provide access to a library of games for online game play. In some examples, code for executing at least some of the games may initially be stored on one or more of the storage devices **282***a*. The code may be subsequently loaded onto a server **284***a* after selection by a player via an EUD and communication of that selection from the EUD via the networks **417**. The server **284***a* onto which code for the selected game has been loaded may provide the game according to selections made by a player and indicated via the player's EUD. In other examples, code for executing at least some of the games may initially be stored on one or more of the servers **284***a*. Although only one gaming data center **276** is shown in FIG. **2**C, some implementations may include multiple gaming data centers **276**.

[0083] In this example, a financial institution data center **270** is also configured for communication via the networks **417**. Here, the financial institution data center **270** includes servers **284***b*, storage devices **282***b*, and one or more workstations **286***b*. According to this example, the financial institution data center **270** is configured to maintain financial accounts, such as checking accounts, savings accounts, loan accounts, etc. In some implementations one or more of the authorized users **274***a*-**274***c* may maintain at least one financial account with the financial institution that is serviced via the financial institution data center **270**.

[0084] According to some implementations, the gaming data center **276** may be configured to provide online wagering games in which money may be won or lost. According to some such implementations, one or more of the servers **284***a* may be configured to monitor player credit balances, which may be expressed in game credits, in currency units, or in any other appropriate manner. In some implementations, the server(s) **284***a* may be configured to obtain financial credits from and/or provide financial credits to one or more financial institutions, according to a player's "cash in" selections, wagering game results and a player's "cash out" instructions. According to some such implementations, the server(s) **284***a* may be configured to electronically credit or debit the account of a player that is maintained by a financial institution, e.g., an account that is maintained via the financial institution data center **270**. The server(s) **284***a* may, in some examples, be configured to maintain an audit record of such transactions.

[0085] In some alternative implementations, the gaming data center **276** may be configured to provide online wagering games for which credits may not be exchanged for cash or the equivalent. In some such examples, players may purchase game credits for online game play, but may not "cash out" for monetary credit after a gaming session. Moreover, although the financial institution data center **270**

and the gaming data center **276** include their own servers and storage devices in this example, in some examples the financial institution data center **270** and/or the gaming data center **276** may use offsite "cloud-based" servers and/or storage devices. In some alternative examples, the financial institution data center **270** and/or the gaming data center **276** may rely entirely on cloud-based servers.

[0086] One or more types of devices in the gaming data center **276** (or elsewhere) may be capable of executing middleware, e.g., for data management and/or device communication. Authentication information, player tracking information, etc., including but not limited to information obtained by EUDs **264** and/or other information regarding authorized users of EUDs **264** (including but not limited to the authorized users **274***a*-**274***c*), may be stored on storage devices **282** and/or servers **284**. Other game-related information and/or software, such as information and/or software relating to leaderboards, players currently playing a game, game themes, game-related promotions, game competitions, etc., also may be stored on storage devices **282** and/or servers **284**. In some implementations, some such game-related software may be available as "apps" and may be downloadable (e.g., from the gaming data center **276**) by authorized users.

[0087] In some examples, authorized users and/or entities (such as representatives of gaming regulatory authorities) may obtain gaming-related information via the gaming data center **276**. One or more other devices (such EUDs **264** or devices of the gaming data center **276**) may act as intermediaries for such data feeds. Such devices may, for example, be capable of applying data filtering algorithms, executing data summary and/or analysis software, etc. In some implementations, data filtering, summary and/or analysis software may be available as "apps" and downloadable by authorized users.

[0088] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture **300** that implements a game processing pipeline for the play of a game in accordance with various implementations described herein. As shown in FIG. **3**, the gaming processing pipeline starts with having a UI system **302** receive one or more player inputs for the game instance. Based on the player input(s), the UI system **302** generates and sends one or more RNG calls to a game processing backend system **314**. Game processing backend system **314** then processes the RNG calls with RNG engine **316** to generate one or more RNG outcomes. The RNG outcomes are then sent to the RNG conversion engine **320** to generate one or more game outcomes for the UI system **302** to display to a player. The game processing architecture **300** can implement the game processing pipeline using a gaming device, such as gaming devices **104**A-**104**X and **200** shown in FIGS. **1** and **2**, respectively. Alternatively, portions of the gaming processing architecture **300** can implement the game processing pipeline using a gaming device and one or more remote gaming devices, such as central determination gaming system server **106** shown in FIG. **1**.

[0089] The UI system **302** includes one or more UIs that a player can interact with. The UI system **302** could include one or more game play UIs **304**, one or more bonus game play UIs **308**, and one or more multiplayer UIs **312**, where each UI type includes one or more mechanical UIs and/or graphical UIs (GUIs). In other words, game play UI **304**, bonus game play UI **308**, and the multiplayer UI **312** may utilize a variety of UI elements, such as mechanical UI elements (e.g., physical "spin" button or mechanical reels) and/or GUI elements (e.g., virtual reels shown on a video display or a virtual button deck) to receive player inputs and/or present game play to a player. Using FIG. **3** as an example, the different UI elements are shown as game play UI elements **306**A-**306**N and bonus game play UI elements **310**A-**310**N.

[0090] The game play UI **304** represents a UI that a player typically interfaces with for a base game. During a game instance of a base game, the game play UI elements **306**A-**306**N (e.g., GUI elements depicting one or more virtual reels) are shown and/or made available to a user. In a subsequent game instance, the UI system **302** could transition out of the base game to one or more bonus games. The bonus game play UI **308** represents a UI that utilizes bonus game play UI elements **310**A-**310**N for a player to interact with and/or view during a bonus game. In one or more implementations, at least some of the game play UI element **306**A-**306**N are similar to the bonus game play UI elements **310**A-**310**N. In other implementations, the game play UI element **306**A-**306**N can differ from the bonus game play UI elements **310**A-**310**N.

[0091] FIG. **3** also illustrates that UI system **302** could include a multiplayer UI **312** purposed for game play that differs or is separate from the typical base game. For example, multiplayer UI **312** could be set up to receive player inputs and/or presents game play information relating to a tournament mode. When a gaming device transitions from a primary game mode that presents the base game to a tournament mode, a single gaming device is linked and synchronized to other gaming devices to generate a tournament outcome. For example, multiple RNG engines **316** corresponding to each gaming device could be collectively linked to determine a tournament outcome. To enhance a player's gaming experience, tournament mode can modify and synchronize sound, music, reel spin speed, and/or other operations of the gaming devices according to the tournament game play. After tournament game play ends, operators can switch back the gaming device from tournament mode to a primary game mode to present the base game. Although FIG. **3** does not explicitly depict that multiplayer UI **312** includes UI elements, multiplayer UI **312** could also include one or more multiplayer UI elements.

[0092] Based on the player inputs, the UI system **302** could generate RNG calls to a game processing backend system **314**. As an example, the UI system **302** could use one or more application programming interfaces (APIs) to generate the RNG calls. To process the RNG calls, the RNG engine **316** could utilize gaming RNG **318** and/or non-gaming RNGs **319**A-**319**N. Gaming RNG **318** could corresponds to RNG **212** or hardware RNG **244** shown in FIG. **2**A. As previously discussed with reference to FIG. **2**A, gaming RNG **318** often performs specialized and non-generic operations that comply with regulatory and/or game requirements. For example, because of regulation requirements, gaming RNG **318** could correspond to RNG **212** by being a cryptographic RNG or pseudorandom number generator (PRNG) (e.g., Fortuna PRNG) that securely produces random numbers for one or more game features. To securely generate random numbers, gaming RNG **318** could collect random data from various sources of entropy, such as from an operating system (OS) and/or a hardware RNG (e.g., hardware RNG **244** shown in FIG. **2**A). Alternatively, non-gaming RNGs **319**A-**319**N may not be cryptographically secure and/or be computationally less expensive. Non-gaming RNGs **319**A-**319**N can, thus, be used to generate outcomes for non-gaming purposes. As an example, non-gaming RNGs **319**A-**319**N can generate random numbers for generating random messages that appear on the gaming device.

[0093] The RNG conversion engine **320** processes each RNG outcome from RNG engine **316** and converts the RNG outcome to a UI outcome that is feedback to the UI system **302**. With reference to FIG. **2**A, RNG conversion engine **320** corresponds to RNG conversion engine **210** used for game play. As previously described, RNG conversion engine **320** translates the RNG outcome from the RNG **212** to a game outcome presented to a player. RNG conversion engine **320** utilizes one or more lookup tables **322**A-**322**N to regulate a prize payout amount for each RNG outcome and how often the gaming device pays out the derived prize payout amounts. In one example, the RNG conversion engine **320** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. In this example, the mapping between the RNG outcome and the game outcome controls the frequency in hitting certain prize payout amounts. Different lookup tables could be utilized depending on the different game modes, for example, a base game versus a bonus game.

[0094] After generating the UI outcome, the game processing backend system **314** sends the UI outcome to the UI system **302**. Examples of UI outcomes are symbols to display on a video reel or reel stops for a mechanical reel. In one example, if the UI outcome is for a base game, the UI system **302** updates one or more game play UI elements **306**A-**306**N, such as symbols, for the game play UI **304**. In another example, if the UI outcome is for a bonus game, the UI system could update one or more bonus game play UI elements **310**A-**310**N (e.g., symbols) for the bonus game play UI **308**. In response to updating the appropriate UI, the player may subsequently provide additional player inputs to initiate a subsequent game instance that progresses through the game processing pipeline.

[0095] FIG. **4** illustrates an example computer system **400** diagram for dynamic theme selection in electronic gaming. For example, system **400** and the additional and/or alternative examples shown in FIGS. **5**A-**11** may be implemented by any of the gaming devices (e.g., **104**A-X, **200**, **256**, **264***a-c*), systems, and/or architectures described herein. In the example shown in FIG. **4**, system **400** includes a game component **402** and a host component **404**. Game component **402** includes a theme selection component **406**, a player favorite component **408**, a theme manager component **410**, and a game client component **412** including a game initialization component **414**. Host component **404** includes a client context component **416**, and a game manager component **418** including a fast game reset component **420**.

[0096] In the example embodiment, a theme selection page (e.g., or plurality of scrollable selection pages with previous, exit and next buttons) including a plurality of themes is caused to be displayed at an electronic gaming device by theme selection component **406**. Upon receiving selection of a theme, theme manager component **410** requests host component **404** reload a game with the selected theme. For example, theme manager component **410** transmits a request **422** including a theme ID associated with the selected theme to client context component **416**.

[0097] In some embodiments, themes may be selected in a number of ways different from selection at a menu, including automatic selection of a theme stored in a player account based upon detection of a player associated with the player account at the gaming device (e.g., based on a card-in event and/or communication with a player device associated with the player account). In other words, player favorite component **408** may automatically determine a player selected theme based on data associated with a player account transmitted from another device, such as player tracking server **110** or a player device (e.g., EUDs **264***a-c*), and theme manager component **410** may transmit request **422** based on a theme ID associated with the player selected theme as determined by player favorite component **408**. Thus, the player may select a theme for storage in their player account such that upon their arrival at a gaming device, the gaming device automatically updates to correspond to the theme stored in the player account.

[0098] Based on request **422**, client context component **416** communicates with game manager component **418** to coordinate a game reset (e.g., from the default theme to the selected theme). Fast game reset component **420** recreates the game with the selected theme and transmits a message **424** to game client component **412** including the recreated game.

[0099] Based on receipt of the recreated game, game initialization component **414** initializes the recreated game on the gaming device and game client component **412** executes the game with the selected theme.

[0100] FIG. **5**A illustrates an example operator selection interface **500** or screenshot showing an output after for dynamic theme selection in electronic gaming. As shown in FIG. **5**A, interface **500** includes instructions **501**, a plurality of themes **502**, including theme **504**, theme **506**, and theme **508**, and a plurality of theme controls **510**, including control **512**, control **514**, and control **516**, each control **512-516** corresponding to one of themes **504-508**.

[0101] Instructions **501** indicate to an operator which selections are available on interface **500**. For example, as shown in FIG. **5**A, instructions include "Select Favorite Team(s)" and "Game will default to one of the favorite team." Accordingly, the operator, in this particular example, can select any number of favorite football teams (e.g., or conferences, divisions, players, coaches, player position groups, etc.), as the example embodiment described herein corresponds to professional football. Notably, it should be appreciated that additional/alternative themes are envisioned and the themes described herein are not limited to football teams, or other sports themes for that matter, and may include any theme (e.g., holidays, movies, musical groups, etc.).

[0102] In the example of FIG. **5**A, three themes **504-508** corresponding to three teams are shown, but all of the teams may be selectable in different areas (not shown) of interface **500**. In the example embodiment, a default selection at controls **512-516** is "No" (e.g., not a selected default). If no theme is selected, at least one theme from the list of themes may be randomly selected. If one theme is selected, that theme is set as the default theme for the game. If more than one theme is selected, the selected themes may rotate and/or alternate. For example, if more than one theme is selected the default themes may rotate at predetermined times (e.g., after a number of plays of a game, a predetermined amount of time, etc.). In some embodiments, default themes may be automatically updated based events, such as events on a calendar. In various embodiments, the list of plurality of themes **502** is dynamically generated and displayed based on certain metadata that is stored in the memory of the gaming device.

[0103] For example, continuing the football theme described herein, a default theme may correspond to an football team in a geographical location in/near a gaming establishment (e.g., in Las Vegas, the Las Vegas (LV) football team may be the default theme). Further, one or more themes may be updated based on the football schedule. For instance, continuing the Las Vegas example above, if the LV football team has an upcoming game against a certain team, that opposing team may be selected as a default team (e.g., in addition to or alternatively from the LV football team). For example, say the Denver football team have an upcoming game against the LV football team in Las Vegas. Accordingly, the default theme may be updated to include both the LV football team and the Denver football team at gaming establishments in Las Vegas (e.g., because fans of the Denver football team may be travelling to Las Vegas for the game and would be better entertained by a default theme including their favorite team). Of course, regardless of the selection of default themes (e.g., by an operator), a player would have the option to change the theme, as described herein in further detail. Accordingly, certain technical benefits are provided herein including dynamic determination of a default theme based upon location and/or time (e.g., a schedule), as examples.

[0104] In another football example, certain themes may be selected to correspond to certain football-related calendar events. For example, during primetime games (e.g., Sunday night football, Monday night football), themes according to those games (e.g., a theme for the primetime game, themes for teams in the game, etc.). In some embodiments, game themes may correspond to other events on a calendar such as holidays (e.g., Halloween in October, Thanksgiving in November, etc.).

[0105] Theme selections, as shown in FIG. **5**A, can be made at any time by an operator and do not require an entire clear of memory of a game. For example, change of a default theme by an operator may not effect a game currently being played (e.g., the default theme change may take place, if a gaming device is in use, after a current gaming session at the gaming device ends). Then, upon a player beginning a new gaming session, a theme selection screen (e.g., see FIG. **7**A) may be presented to allow the player to select a different theme, if they so desire.

[0106] FIG. **5**B illustrates another example screenshot and/or operator selection interface **550** for dynamic theme selection in electronic gaming. For example, similar to plurality of themes **502**, interface **550** includes a plurality of themes **552** (e.g., in this example, corresponding to a plurality of football teams). Further, similar to plurality of controls **510**, interface **550** includes a plurality of selectable controls **554** and a selector **556** operable to be selected to allow for scrolling of plurality of themes **552**.

[0107] FIG. **6** illustrates an example interface **600** or screenshot showing an output after for dynamic theme selection in electronic gaming, in accordance with the present disclosure. As shown in FIG. **6**, interface **600** includes a theme background **601**, theme symbols **602**, a theme selector **604** including a theme indicator **606**.

[0108] Continuing the football example described herein, background **601** includes a football field. Theme symbols **602** are displayed according to a default theme, here being a Dallas football team (e.g., an professional football team, or "Team **5**"). Although symbols **602** are shown in FIG. **6** as being the same theme symbol (e.g., a football helmet), it should be appreciated that theme symbols **602** could be any symbols associated with the theme (e.g., football players, coaches, jerseys, etc.).

[0109] Interface **600** could support play of an electronic game including the selected theme if a player desires to play with the default theme (e.g., spins of the shown reels with at least some symbols associated with the default theme). Additionally/alternatively, a player could select theme selector **604** to select a different theme (e.g., see FIG. **7**A). As shown in FIG. **6**, theme selector **604** includes a theme indicator **606** indicating a currently-selected theme.

[0110] FIG. **7**A illustrates an example selection interface **700** or screenshot showing an output after for dynamic theme selection in electronic gaming, continuing the example shown in FIG. **6** (e.g., displayed upon selection of theme selector **604**). Interface **700** includes a list **702** of available themes (e.g., overlaid upon interface **600**). Continuing the football theme described herein, list **702** includes a plurality of selectors **704** each corresponding to an available theme. Instructions **706** indicate that a player should "Pick Your Team," if so desired. Accordingly, a selection of selector **708**, of the plurality of selectors **704**, is illustrated. Because selector **708** is selected, display of selector **708** is modified (e.g., enlarged, animated, including a colored border/shadow according to selector **708**). Based on selection of selector **708**, the theme of the game will be updated according to the selected theme (e.g., Team **7**), as described herein.

[0111] Although list **702** is illustrated as being displayed all at once, in some embodiments a greater number of theme options may be provided. Accordingly, selectors **704** may be divided into different pages with previous ("PREV"), exit ("EXIT"), and next ("NEXT") selectors provided such that a player can navigate to a previous page of selectors **704**, exit list **702** (e.g., such that list **702** is no longer overlaid upon interface **600**), or navigate to a next page of selectors **704** respectively.

[0112] FIG. **7**B illustrates another example selection screenshot and/or interface **750** for dynamic theme selection in electronic gaming, continuing the example shown in FIG. **6**. For instance, similar to FIG. **7**A, interface **750** includes a list **752** of available themes including a plurality of selectors **754**. Further, a scrollable list **756**, corresponding to plurality of selectors **754**, is provided. For example, using a touchscreen of interface **750**, a player could "swipe" through list **756** to select a selector in list **756** and/or select from plurality of selectors **754** shown in list **752**. To select the same theme shown as selected in FIG. **7**A, a player may select a selector "7" associated with Team **7**.

[0113] As shown in FIG. **7**B, plurality of selectors **754** are divided by a sub-theme (e.g., here, football conference such as Conference "A" or Conference "N"). Notably, selectors **754** corresponding to any theme may be displayed and/or sub-divided in any was deemed fit (e.g., by a game provider).

[0114] FIG. **7**C illustrates an example transition screenshot and/or interface **780**, continuing the examples shown in

FIGS. **7**A and/or **7**B. For example, upon selection of selector **708** (as shown in FIG. **7**A) or a selector in FIG. **7**B, a transition animation may be displayed including an indication **782** of the selected theme (e.g., as shown in FIG. **7**C, a team logo for Team **7**).

[0115] FIG. **8**A illustrates an example interface **800** or screenshot showing an output after for dynamic theme selection in electronic gaming, based on the selection of selector **708** as shown in FIG. **7**A or a selector in FIG. **7**B. Interface **800** includes a theme background **801** (e.g., matching background **601**, as the theme still corresponds to football), a plurality of theme symbols **802** corresponding to the selection of selector **708** (e.g., and replacing symbols **602**, as shown in FIG. **6**), and a theme selector **804** (e.g., as shown in FIG. **6**) including a theme indicator **806** (e.g., as updated from indicator **606** shown in FIG. **6**).

[0116] For example, with respect to FIGS. **4**, **7**A, and **8**A, list **702** including a plurality of themes is displayed as controlled by theme selection component **406**. Upon receiving selection of selector **708**, theme manager component **410** requests host component **404** reload a game with the selected theme (e.g., theme manager component **410** transmits a request **422** including a theme ID associated with selector **708** to client context component **416**).

[0117] Based on request **422**, client context component **416** communicates with game manager component **418** to coordinate a game reset (e.g., from the default theme, Team **5**, to the selected theme, Team **7**). Fast game reset component **420** recreates the game according to selection of selector **708** and transmits a message **424** to game client component **412** including the recreated game.

[0118] Based on receipt of the recreated game, game initialization component **414** initializes the recreated game on the gaming device (e.g., causes display of interface **800**) and game client component **412** executes the game with the selected theme. The electronic game is displayed with the selected theme including theme symbols **802**. A game outcome, including at least one symbol of theme symbols **802** (e.g., or other theme symbols associated with the theme), for the electronic game is generated, displayed, and stored in a game log as being associated with the identifier. Thus, upon replay and/or recovery of the game outcome, the replay and/or recovery includes the game outcome including the at least one symbol based on the identifier.

[0119] FIG. **8**B illustrates another example screenshot and/or interface **850** showing an output after for dynamic theme selection in electronic gaming, based on a selection made in FIG. **7**B. For instance, similar to FIG. **8**A, interface **850** includes a theme background **851**, a plurality of theme symbols **852** corresponding to the selection of a selector of list **756** (e.g., Team **7**), and a theme selector **854** including a theme indicator **856**. An additional theme indicator **858** corresponding to the selected theme is also shown in FIG. **8**B.

[0120] FIG. **9** illustrates an example screenshot and/or interface **900** illustrating a first theme for dynamic theme selection in electronic gaming, in accordance with the present disclosure. In the example shown in FIG. **9**, a player-selected theme (e.g., Team **25**) is shown. For example, symbols **902** are tailored to the player-selected them as well as other aspects of the game (e.g., sounds). A theme selector **904** is displayed including a theme indicator **906** corresponding to the player-selected theme. An additional theme indicator **908** is also displayed as corresponding to the player-selected theme.

[0121] FIG. **10** illustrates an example screenshot and/or interface **1000** illustrating a transition to an operator-selected theme, continuing the example shown in FIG. **9**. For example, as explained herein, after a predetermined amount of inactivity at an electronic gaming device, the electronic gaming device may revert to one or more default themes (e.g., selected by an operator). In some embodiments, themes of games may automatically be changed for reasons other than inactivity, as explained elsewhere herein (e.g., occurrence of a real-time event, a schedule, etc.). As shown in FIG. **10**, a transition animation including a new theme indicator **1002** (e.g., corresponding to Team **9**) is displayed because of one or more default themes being Team **9**. As explained elsewhere herein, default themes may rotate if more than one default theme is selected. Accordingly, the animation shown in FIG. **10** may be displayed with each such transition between themes (e.g., including a respective theme indicator).

[0122] FIG. **11** illustrates an example screenshot and/or interface **1100** illustrating an operator-selected theme, continuing the example shown in FIG. **10**. For example, as shown in FIG. **11**, the game of FIG. **9** is now displayed with the default theme (e.g., shown in FIG. **10**) instead of the player-selected theme. Here, as illustrated in FIG. **10**, a default theme corresponds to Team **9**. Accordingly, symbols **1102** have changed to correspond to the theme of Team **9** (e.g., rather than the theme of Team **25**, as shown in FIG. **9**), while remaining in the symbol positions shown in FIG. **9** (e.g., because a next play of the game has not yet occurred). A theme selector **1104** is displayed including a theme indicator **1106** corresponding to the player-selected theme. An additional theme indicator **1108** is also displayed as corresponding to the player-selected theme.

[0123] FIG. **12** illustrates an example method **1200** for dynamic theme selection in electronic gaming, in accordance with the present disclosure. In the example embodiment, method **1200** includes causing display **1202** of an electronic game with a first theme (e.g., Team **5**), the first theme including a first plurality of symbols (e.g., **602**), receiving **1204** an input associated with a second theme (e.g., Team **7**), and transmitting **1206** an identifier associated with the second theme to a host computer component (e.g., in some embodiments, the host component is a component of a processor and in some embodiments, the host is code executed by a processor). Method **1200** also includes receiving **1208** content (e.g., including symbols, an animation, a video, a background animation, a voice over, a reel background, and/or a help screen) associated with the second theme from the host computer component wherein the content includes a second plurality of symbols (e.g., **802**) and causing display **1210** of the electronic game with the second theme, the second theme including the second plurality of symbols. Method **1200** further includes causing **1212** a game outcome for the electronic game to be stored in a game log in at least one memory as being associated with the identifier wherein the game outcome includes at least one symbol of the second plurality of symbols, and wherein upon generation of a replay of the game outcome, the replay includes the at least one symbol based on the identifier and the game outcome and causing display **1214** of the game outcome.

[0124] In some embodiments, method **1200** includes causing display of a menu including a plurality of selectable icons associated with a plurality of themes, the plurality of themes including the first theme and the second theme and receiving the input associated with the second theme at the menu.

[0125] In some embodiments, method **1200** includes identifying a default theme stored in memory and causing display of the electronic game with the first theme, wherein the first theme includes the default theme. As an example, in some embodiments the default theme may include a plurality of themes. In some embodiments, method **1200** includes determining that the default theme stored in memory has changed to an updated default theme (e.g., based on a schedule, an operator input, etc.) and causing display of the electronic game with the updated default theme.

[0126] In some embodiments, method **1200** includes determining that a game session of the electronic game with the second theme has ended and based on the game session ending, causing display of the electronic game with the first theme. In some embodiments, method **1200** includes receiving the input associated with the second theme from an end user device associated with a player wherein the input identifies the second theme as being selected by the player and stored in a player account associated with the player.

[0127] FIG. **13** illustrates another example method **1300** for dynamic theme selection in electronic gaming, in accordance with the present disclosure. In the example embodiment, method **1300** includes storing **1302** a plurality of themes in memory, receiving **1304** an input associated with a theme of the plurality of themes, and transmitting **1306** an identifier associated with the theme to a host component of the at least one processor. Further, method **1300** includes receiving **1308** content associated with the theme from the host component wherein the content includes a plurality of symbols (e.g., and/or an animation, a video, a background animation, a voice over, a reel background, and/or a help screen) and causing display **1310** of an electronic game with the theme, the theme including the plurality of symbols. Method **1300** also includes causing **1312** a game outcome for the electronic game to be stored in a game log in memory as being associated with the identifier wherein the game outcome includes at least one symbol of the plurality of symbols, and wherein upon generation of a replay of the game outcome, the replay includes the at least one symbol based on the identifier and the game outcome and causing display **1314** of the game outcome.

[0128] In some embodiments, method **1300** includes causing display of a menu including a plurality of selectable icons associated with a plurality of themes, the plurality of themes including the theme and receiving the input associated with the theme at the menu. In some embodiments, method **1300** includes determining that a game session of the electronic game with the theme has ended and, based on the game session ending, causing display of the electronic game with a default theme.

[0129] While the disclosure has been described with respect to the figures, it will be appreciated that many modifications and changes may be made by those skilled in the art without departing from the spirit of the disclosure. Any variation and derivation from the above description and figures are included in the scope of the present disclosure as defined by the claims.

What is claimed is:

1. An electronic gaming system comprising:

at least one memory with instructions stored thereon; and

at least one processor in communication with the at least one memory, wherein the instructions, when executed by the at least one processor, cause the at least one processor to:

transmit a first request to update an electronic game to a host component of the electronic gaming system based upon receiving an input associated with selecting a first theme from a plurality of themes, the first request comprising an identifier associated with the first theme;

receive updated game data associated with the first theme from the host component based upon the first request, the updated game data generated based upon the first request and comprising a first plurality of symbols associated with the first theme to replace placeholder symbols during play of the electronic game;

cause a game outcome to be provided for the electronic game, the game outcome including at least one symbol of the first plurality of symbols replacing at least one placeholder symbol of the placeholder symbols;

cause the game outcome for the electronic game to be stored in a game log in the at least one memory as including the at least one placeholder symbol and being associated with the identifier, wherein storage of the game outcome does not include storage of the at least one symbol;

receive a second request to one of replay or restore the electronic game;

retrieve the game outcome from the at least one memory based upon the second request;

retrieve the at least one symbol from the host component based upon the identifier; and

cause display of the game outcome as including the at least one symbol replacing the at least one placeholder symbol based upon the identifier stored in the game log as being associated with the game outcome such that the replayed or restored game outcome matches the provided game outcome.

2. The electronic gaming system of claim **1**, wherein the instructions further cause the at least one processor to initialize and execute the electronic game based upon receiving the updated game data.

3. The electronic gaming system of claim **1**, wherein the instructions further cause the at least one processor to:

cause display of a menu including a plurality of selectable icons associated with the plurality of themes, the plurality of themes including the first theme; and

receive the input associated with the first theme at the menu.

4. The electronic gaming system of claim **1**, wherein the instructions further cause the at least one processor to, before receiving the input associated with the first request, cause display of the electronic game as having a default theme associated with at least one default symbol that is displayed, wherein the default theme is stored in the at least one memory.

5. The electronic gaming system of claim **4**, wherein the instructions further cause the at least one processor to:

determine that the default theme stored in the at least one memory has changed to an updated default theme associated with at least one updated default symbol; and

cause display of the electronic game as having the updated default theme associated with the at least one updated default symbol that is displayed.

6. The electronic gaming system of claim **4**, wherein the instructions further cause the at least one processor to:

determine that a game session of the electronic game with the first theme has ended; and

based on the game session ending, cause display of the electronic game as having the default theme.

7. The electronic gaming system of claim **1**, wherein the updated game data further comprises further comprises at least one of an animation, a video, a background animation, a voice over, a reel background, or a help screen.

8. The electronic gaming system of claim **1**, wherein the instructions further cause the at least one processor to:

transmit the first request from a theme manager component to a client context component of the host component; and

receive the updated game data at a game client component from a game reset component of the host component, wherein the client context component causes the game reset component to transmit the updated game data.

9. At least one non-transitory computer-readable storage medium with instructions stored thereon that, in response to execution by at least one processor, cause the at least one processor to:

transmit a first request to recreate an electronic game to a host component associated with the electronic game based upon receiving an input associated with selecting a first theme from a plurality of themes, the first request comprising an identifier associated with the first theme;

receive recreated game data associated with the first theme from the host component based upon the first request, the recreated game data generated based upon the first request and comprising a first plurality of symbols associated with the first theme to replace placeholder symbols during play of the electronic game;

cause a game outcome to be provided for the electronic game, the game outcome including at least one symbol of the first plurality of symbols replacing at least one placeholder symbol of the placeholder symbols;

cause the game outcome for the electronic game to be stored in a game log in the at least one non-transitory computer-readable storage medium as including the at least one placeholder symbol and being associated with the identifier, wherein storage of the game outcome does not include storage of the at least one symbol;

receive a second request to one of replay or restore the electronic game;

retrieve the game outcome from the at least one non-transitory computer-readable storage medium based upon the second request;

retrieve the at least one symbol from the host component based upon the identifier; and

cause display of the game outcome as including the at least one symbol replacing the at least one placeholder symbol based upon the identifier stored in the game log as being associated with the game outcome such that the replayed or restored game outcome matches the provided game outcome.

10. The at least one non-transitory computer-readable storage medium of claim **9**, wherein the instructions further cause the at least one processor to initialize and execute the electronic game based upon receiving the recreated game data.

11. The at least one non-transitory computer-readable storage medium of claim **9**, wherein the instructions further cause the at least one processor to:

cause display of a menu including a plurality of selectable icons associated with the plurality of themes, the plurality of themes including the first theme; and

receive the input associated with the first theme at the menu.

12. The at least one non-transitory computer-readable storage medium of claim **9**, wherein the instructions further cause the at least one processor to, before receiving the input associated with the first request, cause display of the electronic game as having a default theme associated with at least one default symbol that is displayed, wherein the default theme is stored in the at least one non-transitory computer-readable storage medium.

13. The at least one non-transitory computer-readable storage medium of claim **12**, wherein the instructions further cause the at least one processor to:

determine that the default theme stored in the at least one non-transitory computer-readable storage medium has changed to an updated default theme associated with at least one updated default symbol; and

cause display of the electronic game as having the updated default theme associated with the at least one updated default symbol that is displayed.

14. The at least one non-transitory computer-readable storage medium of claim **12**, wherein the instructions further cause the at least one processor to:

determine that a game session of the electronic game with the first theme has ended; and

based on the game session ending, cause display of the electronic game as having the default theme.

15. The at least one non-transitory computer-readable storage medium of claim **9**, wherein the recreated game data further comprises further comprises at least one of an animation, a video, a background animation, a voice over, a reel background, or a help screen.

16. The at least one non-transitory computer-readable storage medium of claim **9**, wherein the instructions further cause the at least one processor to:

transmit the first request from a theme manager component to a client context component of the host component; and

receive the recreated game data at a game client component from a game reset component of the host component, wherein the client context component causes the game reset component to transmit the recreated game data.

17. A method of dynamic theme selection in electronic gaming, the method implemented by at least one processor in communication with at least one memory, the method comprising:

transmitting a first request to update an electronic game to a host component associated with the electronic game based upon receiving an input associated with selecting a first theme from a plurality of themes, the first request comprising an identifier associated with the first theme;

receiving updated game data associated with the first theme from the host component based upon the first request, the updated game data generated based upon the first request and comprising a first plurality of symbols associated with the first theme to replace placeholder symbols during play of the electronic game;

causing a game outcome to be provided for the electronic game, the game outcome including at least one symbol of the first plurality of symbols replacing at least one placeholder symbol of the placeholder symbols;

causing the game outcome for the electronic game to be stored in a game log in the at least one memory as including the at least one placeholder symbol and being associated with the identifier, wherein storage of the game outcome does not include storage of the at least one symbol;

receiving a second request to one of replay or restore the electronic game;

retrieving the game outcome from the at least one memory based upon the second request;

retrieving the at least one symbol from the host component based upon the identifier; and

causing display of the game outcome as including the at least one symbol replacing the at least one placeholder symbol based upon the identifier stored in the game log as being associated with the game outcome such that the replayed or restored game outcome matches the provided game outcome.

18. The method of claim **17**, further comprising initializing and executing the electronic game based upon receiving the updated game data.

19. The method of claim **17**, further comprising:

before receiving the input associated with the first request, causing display of the electronic game as having a default theme associated with at least one default symbol that is displayed, wherein the default theme is stored in the at least one memory;

determining that the default theme stored in the at least one memory has changed to an updated default theme associated with at least one updated default symbol; and

causing display of the electronic game as having the updated default theme associated with the at least one updated default symbol that is displayed.

20. The method of claim **17**, further comprising:

transmitting the first request from a theme manager component to a client context component of the host component; and

receiving the updated game data at a game client component from a game reset component of the host component, wherein the client context component causes the game reset component to transmit the updated game data.

* * * * *